C has "signed URL", everyone who has the URL could download. Plus, only B ensure the "must be downloaded securely" part by using VPN.
upvoted 5 times

NayeraB Most Recent 1 year, 3 months ago

Selected Answer: B

My money is on B, but it's still not mentioned that the customer used an on-prem Active Directory.
upvoted 3 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: B

Windows file server = Amazon FSx for Windows File Server file system
Files can be accessed only by authorized users = On-premises Active Directory
upvoted 4 times

JA2018 6 months, 2 weeks ago

"Data must be downloaded securely to users' devices" => AWS Client VPN
upvoted 1 times

BrijMohan08 1 year, 9 months ago

Selected Answer: C

Remember: The file server is running out of capacity.
upvoted 2 times

awsgeek75 1 year, 4 months ago

But then how do you download the files to user's machine in a secure way?
upvoted 2 times

pentium75 1 year, 5 months ago

That's why we're using FSX for Windows File Server in AWS.

"Signed URL to allow download" would allow everyone who has the URL to download the files, but we must "ensure that the files can be accessed only by authorized users". Plus, the "private VPC endpoint" is not really of use here, it's still S3 and the users are not in AWS.
upvoted 4 times

SkyZeroZx 2 years, 1 month ago

Selected Answer: B

B is the correct answer
upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: B

B is the best solution for the given requirements. It provides a secure way for employees to access confidential and sensitive files from anywhere using AWS Client VPN. The Amazon FSx for Windows File Server file system is designed to provide native support for Windows file system features such as NTFS permissions, Active Directory integration, and Distributed File System (DFS). This means that the company can continue to use their on-premises Active Directory to manage user access to files.
upvoted 4 times

Bilalazure 2 years, 3 months ago

B is the correct answer
upvoted 2 times

jennyka76 2 years, 3 months ago

Answer - B
1- https://docs.aws.amazon.com/fsx/latest/WindowsGuide/what-is.html
2- https://docs.aws.amazon.com/fsx/latest/WindowsGuide/managing-storage-capacity.html
upvoted 2 times

Neha999 2 years, 3 months ago

B
Amazon FSx for Windows File Server file system
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #333 *Topic 1*

A company's application runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. On the first day of every month at midnight, the application becomes much slower when the month-end financial calculation batch runs. This causes the CPU utilization of the EC2 instances to immediately peak to 100%, which disrupts the application.

What should a solutions architect recommend to ensure the application is able to handle the workload and avoid downtime?

A. Configure an Amazon CloudFront distribution in front of the ALB.

B. Configure an EC2 Auto Scaling simple scaling policy based on CPU utilization.

C. Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule. Most Voted

D. Configure Amazon ElastiCache to remove some of the workload from the EC2 instances.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

TariqKipkemei Highly Voted 8 months ago

Selected Answer: C

'On the first day of every month at midnight' = Scheduled scaling policy

upvoted 5 times

elearningtakai Most Recent 1 year, 2 months ago

Selected Answer: C

By configuring a scheduled scaling policy, the EC2 Auto Scaling group can proactively launch additional EC2 instances before the CPU utilization peaks to 100%. This will ensure that the application can handle the workload during the month-end financial calculation batch, and avoid any disruption or downtime.

Configuring a simple scaling policy based on CPU utilization or adding Amazon CloudFront distribution or Amazon ElastiCache will not directly address the issue of handling the monthly peak workload.

upvoted 4 times

Steve_4542636 1 year, 3 months ago

Selected Answer: C

If the scaling were based on CPU or memory, it requires a certain amount of time above that threshhold, 5 minutes for example. That would mean the CPU would be at 100% for five minutes.

upvoted 4 times

LuckyAro 1 year, 3 months ago

Selected Answer: C

C: Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule is the best option because it allows for the proactive scaling of the EC2 instances before the monthly batch run begins. This will ensure that the application is able to handle the increased workload without experiencing downtime. The scheduled scaling policy can be configured to increase the number of instances in the Auto Scaling group a few hours before the batch run and then decrease the number of instances after the batch run is complete. This will ensure that the resources are available when needed and not wasted when not needed.

The most appropriate solution to handle the increased workload during the monthly batch run and avoid downtime would be to configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule.

upvoted 3 times

LuckyAro 1 year, 3 months ago

Scheduled scaling policies allow you to schedule EC2 instance scaling events in advance based on a specified time and date. You can use this feature to plan for anticipated traffic spikes or seasonal changes in demand. By setting up scheduled scaling policies, you can ensure that you have the right number of instances running at the right time, thereby optimizing performance and reducing costs.

To set up a scheduled scaling policy in EC2 Auto Scaling, you need to specify the following:

Start time and date: The date and time when the scaling event should begin.

Desired capacity: The number of instances that you want to have running after the scaling event.

Recurrence: The frequency with which the scaling event should occur. This can be a one-time event or a recurring event, such as daily or weekly.

upvoted 2 times

bdp123 1 year, 3 months ago

Selected Answer: C

C is the correct answer as traffic spike is known

upvoted 2 times

jennyka76 1 year, 3 months ago

ANSWER - C
https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scheduled-scaling.html

upvoted 3 times

Neha999 1 year, 3 months ago

C as the schedule of traffic spike is known beforehand.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #334 *Topic 1*

A company wants to give a customer the ability to use on-premises Microsoft Active Directory to download files that are stored in Amazon S3. The customer's application uses an SFTP client to download the files.

Which solution will meet these requirements with the LEAST operational overhead and no changes to the customer's application?

A. Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication. Most Voted

B. Set up AWS Database Migration Service (AWS DMS) to synchronize the on-premises client with Amazon S3. Configure integrated Active Directory authentication.

C. Set up AWS DataSync to synchronize between the on-premises location and the S3 location by using AWS IAM Identity Center (AWS Single Sign-On).

D. Set up a Windows Amazon EC2 instance with SFTP to connect the on-premises client with Amazon S3. Integrate AWS Identity and Access Management (IAM).

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

Steve_4542636 Highly Voted 2 years, 3 months ago

Selected Answer: A

SFTP, FTP - think "Transfer" during test time

upvoted 17 times

wsdasdasdqwdaw Most Recent 1 year, 7 months ago

LEAST operational overhead => A, D is much more operational overhead

upvoted 3 times

JA2018 6 months, 2 weeks ago

1. AWS Transfer Family: Fully managed service that allows customers to transfer files over SFTP, FTPS, and FTP directly into and out of Amazon S3.

2. Eliminates the need to manage any infrastructure for file transfer, which reduces operational overhead.

3. You can also configure the service to use an existing Active Directory for authentication, =>>> no changes need to be made to the customer's application.

4. Rule of thumbs for AWS exams: Opt for AWS fully managed services. :p :p :p

upvoted 1 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

SFTP, No changes to the customer's application? = AWS Transfer Family

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Transfer family is used for SFTP

upvoted 2 times

live_reply_developers 1 year, 10 months ago

SFTP -> transfer family

upvoted 2 times

antropaws 2 years ago

Selected Answer: A

A no doubt. Why the system gives B as the correct answer?

upvoted 2 times

lht 2 years, 1 month ago

Selected Answer: A

just A

upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: A

AWS Transfer Family

upvoted 3 times

LuckyAro 2 years, 3 months ago

AWS Transfer Family is a fully managed service that allows customers to transfer files over SFTP, FTPS, and FTP directly into and out of Amazon S3. It eliminates the need to manage any infrastructure for file transfer, which reduces operational overhead. Additionally, the service can be configured to use an existing Active Directory for authentication, which means that no changes need to be made to the customer's application.

upvoted 3 times

bdp123 2 years, 3 months ago

Selected Answer: A

Transfer family is used for SFTP

upvoted 2 times

TungPham 2 years, 3 months ago

Selected Answer: A

using AWS Batch to LEAST operational overhead
and have SFTP to no changes to the customer's application

https://aws.amazon.com/vi/blogs/architecture/managed-file-transfer-using-aws-transfer-family-and-amazon-s3/

upvoted 3 times

Bhawesh 2 years, 3 months ago

Selected Answer: A

A. Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication.

https://docs.aws.amazon.com/transfer/latest/userguide/directory-services-users.html
upvoted 4 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #335 *Topic 1*

A company is experiencing sudden increases in demand. The company needs to provision large Amazon EC2 instances from an Amazon Machine Image (AMI). The instances will run in an Auto Scaling group. The company needs a solution that provides minimum initialization latency to meet the demand.

Which solution meets these requirements?

A. Use the aws ec2 register-image command to create an AMI from a snapshot. Use AWS Step Functions to replace the AMI in the Auto Scaling group.

B. Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI. Most Voted

C. Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager (Amazon DLM). Create an AWS Lambda function that modifies the AMI in the Auto Scaling group.

D. Use Amazon EventBridge to invoke AWS Backup lifecycle policies that provision AMIs. Configure Auto Scaling group capacity limits as an event source in EventBridge.

Correct Answer: *B*

Community vote distribution

B (94%) Other

Comments

danielklein09 Highly Voted 2 years ago

readed the question 5 times, didn't understood a thing :(

upvoted 63 times

elmyth 8 months, 3 weeks ago

Me too(((terrible question

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Me too

upvoted 4 times

lostmagnet001 1 year, 4 months ago

the same here!
upvoted 1 times

bdp123 Highly Voted 2 years, 3 months ago

Selected Answer: B

Enabling Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot allows you to quickly create a new Amazon Machine Image (AMI) from a snapshot, which can help reduce the initialization latency when provisioning new instances. Once the AMI is provisioned, you can replace the AMI in the Auto Scaling group with the new AMI. This will ensure that new instances are launched from the updated AMI and are able to meet the increased demand quickly.
upvoted 14 times

LeonSauveterre Most Recent 6 months, 1 week ago

Selected Answer: B

The question focuses on reducing initialization latency when launching large EC2 instances in an Auto Scaling group during sudden demand spikes.

So B with fast snapshot is obviously the choice. But without the choices listed there, I would simply integrate a Warm Pool for the Auto Scaling group with pre-initialized instances based on an optimized AMI :)
upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

The question wording is pretty weird but the only thing of value is latency during initialisation which makes B the correct option.
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

A only helps with creating the AMI
C and D will probably work (ambiguous language) but won't handle initialising latency issues.
upvoted 6 times

farnamjam 1 year, 5 months ago

Selected Answer: B

Fast Snapshot Restore (FSR)
• Force full initialization of snapshot to have no
latency on the first use
upvoted 2 times

pentium75 1 year, 5 months ago

Selected Answer: B

"Fast snapshot restore" = pre-warmed snapshot
AMI from such a snapshot is pre-warmed AMI
upvoted 5 times

master9 1 year, 5 months ago

Selected Answer: D

Amazon Data Lifecycle Manager (DLM) is a feature of Amazon EBS that automates the creation, retention, and deletion of snapshots, which are used to back up your Amazon EBS volumes. With DLM, you can protect your data by implementing a backup strategy that aligns with your business requirements.

You can create lifecycle policies to automate snapshot management. Each policy includes a schedule of when to create snapshots, a retention rule with a defined period to retain each snapshot, and a set of Amazon EBS volumes to assign to the policy.

This service helps simplify the management of your backups, ensure compliance, and reduce costs.
upvoted 1 times

pentium75 1 year, 5 months ago

We're not asked to "simplify the management of our backups, ensure compliance, and reduce costs", we're asked to "provide

minimum initialization latency" for an auto-scaling group.
upvoted 2 times

master9 1 year, 5 months ago

Sorry, its "C" and not "D"
upvoted 1 times

Nisarg2121 1 year, 5 months ago

Selected Answer: B

b is correct
upvoted 1 times

meowruki 1 year, 6 months ago

B. Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI.

Here's the reasoning:

Amazon EBS Fast Snapshot Restore: This feature allows you to quickly create new EBS volumes (and subsequently AMIs) from snapshots. Fast Snapshot Restore optimizes the initialization process by pre-warming the snapshots, reducing the time it takes to create volumes from those snapshots.

Provision an AMI using the snapshot: By using fast snapshot restore, you can efficiently provision an AMI from the pre-warmed snapshot, minimizing the initialization latency.

Replace the AMI in the Auto Scaling group: This allows you to update the instances in the Auto Scaling group with the new AMI efficiently, ensuring that the new instances are launched with minimal delay.
upvoted 2 times

meowruki 1 year, 6 months ago

Option A (Use aws ec2 register-image command and AWS Step Functions): While this approach can be used to automate the creation of an AMI and update the Auto Scaling group, it may not offer the same level of optimization for initialization latency as Amazon EBS fast snapshot restore.

Option C (Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager, create a Lambda function): While Amazon DLM can help manage the lifecycle of your AMIs, it might not provide the same level of speed and responsiveness needed for sudden increases in demand.

Option D (Use Amazon EventBridge and AWS Backup): AWS Backup is primarily designed for backup and recovery, and it might not be as optimized for quickly provisioning instances in response to sudden demand spikes. EventBridge can be used for event-driven architectures, but in this context, it might introduce unnecessary complexity.
upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: B

Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI
upvoted 2 times

kambarami 1 year, 8 months ago

Pleaw3 reword 5he question. Can not understand a thing!
upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Enable EBS fast snapshot restore on a snapshot
Create an AMI from the snapshot
Replace the AMI used by the Auto Scaling group with this new AMI

The key points:

° Need to launch large EC2 instances quickly from an AMI in an Auto Scaling group
° Looking to minimize instance initialization latency

upvoted 3 times

antropaws 2 years ago

Selected Answer: B

B most def

upvoted 2 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

B: "EBS fast snapshot restore": minimizes initialization latency. This is a good choice.

upvoted 3 times

Zox42 2 years, 2 months ago

Selected Answer: B

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

upvoted 3 times

geekgirl22 2 years, 3 months ago

Keyword, minimize initilization latency == snapshot. A and B have snapshots in them, but B is the one that makes sense.
C has DLP that can create machines from AMI, but that does not talk about latency and snapshots.

upvoted 4 times

LuckyAro 2 years, 3 months ago

Selected Answer: B

Enabling Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot allows for rapid restoration of EBS volumes from snapshots. This reduces the time required to create an AMI from a snapshot, which is useful for quickly provisioning large Amazon EC2 instances.

Provisioning an AMI by using the fast snapshot restore feature is a fast and efficient way to create an AMI. Once the AMI is created, it can be replaced in the Auto Scaling group without any downtime or disruption to running instances.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #336 *Topic 1*

A company hosts a multi-tier web application that uses an Amazon Aurora MySQL DB cluster for storage. The application tier is hosted on Amazon EC2 instances. The company's IT security guidelines mandate that the database credentials be encrypted and rotated every 14 days.

What should a solutions architect do to meet this requirement with the LEAST operational effort?

A. Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days. **Most Voted**

B. Create two parameters in AWS Systems Manager Parameter Store: one for the user name as a string parameter and one that uses the SecureString type for the password. Select AWS Key Management Service (AWS KMS) encryption for the password parameter, and load these parameters in the application tier. Implement an AWS Lambda function that rotates the password every 14 days.

C. Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system in all EC2 instances of the application tier. Restrict the access to the file on the file system so that the application can read the file and that only super users can modify the file. Implement an AWS Lambda function that rotates the key in Aurora every 14 days and writes new credentials into the file.

D. Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon S3 bucket that the application uses to load the credentials. Download the file to the application regularly to ensure that the correct credentials are used. Implement an AWS Lambda function that rotates the Aurora credentials every 14 days and uploads these credentials to the file in the S3 bucket.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

elearningtakai Highly Voted 2 years, 2 months ago

Selected Answer: A

AWS Secrets Manager allows you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. With this service, you can automate the rotation of secrets, such as database credentials, on a

throughout their lifecycle. With this service, you can automate the rotation of secrets, such as database credentials, on a schedule that you choose. The solution allows you to create a new secret with the appropriate credentials and associate it with the Aurora DB cluster. You can then configure a custom rotation period of 14 days to ensure that the credentials are automatically rotated every two weeks, as required by the IT security guidelines. This approach requires the least amount of operational effort as it allows you to manage secrets centrally without modifying your application code or infrastructure.
upvoted 7 times

jennyka76 Highly Voted 2 years, 3 months ago

Answer is A
To implement password rotation lifecycles, use AWS Secrets Manager. You can rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle using Secrets Manager.
https://aws.amazon.com/blogs/security/how-to-use-aws-secrets-manager-rotate-credentials-amazon-rds-database-types-oracle/
upvoted 5 times

MJ45 Most Recent 5 months, 2 weeks ago

Selected Answer: A

Most questions related to rotating credentials - AWS Secrets Manager
AWS System Parameter Store CANNOT rotate credentials.
upvoted 1 times

LeonSauveterre 6 months, 1 week ago

Selected Answer: A

Well... I would simply rule out all the options with "Implement an AWS Lambda function" given that we need a solution with least operational effort. Why manually compose codes to rotate when you can achieve the same goal automatically?
upvoted 1 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days
upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

Use AWS Secrets Manager to store the Aurora credentials as a secret
Encrypt the secret with a KMS key
Configure 14 day automatic rotation for the secret
Associate the secret with the Aurora DB cluster
The key points:

Aurora MySQL credentials must be encrypted and rotated every 14 days
Want to minimize operational effort
upvoted 3 times

elearningtakai 2 years, 2 months ago

Selected Answer: A

A: AWS Secrets Manager. Simply this supported rotate feature, and secure to store credentials instead of EFS or S3.
upvoted 2 times

Steve_4542636 2 years, 3 months ago

Selected Answer: A

Voting A
upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: A

A proposes to create a new AWS KMS encryption key and use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Then, the secret will be associated with the Aurora DB cluster, and a custom rotation period of 14 days will be configured. AWS Secrets Manager will automate the process of rotating the database credentials, which will reduce the operational effort required to meet the IT security guidelines.

which will reduce the operational effort required to meet the IT security guidelines.
upvoted 2 times

Neha999 2 years, 3 months ago

A
https://www.examtopics.com/discussions/amazon/view/59985-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #337 *Topic 1*

A company has deployed a web application on AWS. The company hosts the backend database on Amazon RDS for MySQL with a primary DB instance and five read replicas to support scaling needs. The read replicas must lag no more than 1 second behind the primary DB instance. The database routinely runs scheduled stored procedures.

As traffic on the website increases, the replicas experience additional lag during periods of peak load. A solutions architect must reduce the replication lag as much as possible. The solutions architect must minimize changes to the application code and must minimize ongoing operational overhead.

Which solution will meet these requirements?

A. Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions. **Most Voted**

B. Deploy an Amazon ElastiCache for Redis cluster in front of the database. Modify the application to check the cache before the application queries the database. Replace the stored procedures with AWS Lambda functions.

C. Migrate the database to a MySQL database that runs on Amazon EC2 instances. Choose large, compute optimized EC2 instances for all replica nodes. Maintain the stored procedures on the EC2 instances.

D. Migrate the database to Amazon DynamoDB. Provision a large number of read capacity units (RCUs) to support the required throughput, and configure on-demand capacity scaling. Replace the stored procedures with DynamoDB streams.

Correct Answer: *A*

Community vote distribution

A (90%) B (10%)

Comments

fkie4 Highly Voted 2 years, 3 months ago

i hate this kind of question

upvoted 62 times

asoli Highly Voted 2 years, 2 months ago

Selected Answer: A

Using Cache required huge changes in the application. Several things need to change to use cache in front of the DB in the

application. So, option B is not correct.
Aurora will help to reduce replication lag for read replica
upvoted 13 times

sheilawu Most Recent 12 months ago

Selected Answer: A

You need to read the question carefully.
The solutions architect must minimize changes to the application code = therefore A
If this question without this statement, B will be a better choice.
upvoted 4 times

JackyCCK 1 year, 2 months ago

minimize ongoing operational overhead = Not B
Using ElastiCache require app change
upvoted 4 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

AWS Aurora and Native Functions are least application changes while providing better performance and minimum latency.
https://aws.amazon.com/rds/aurora/faqs/

B, C, D require lots of changes to the application so relatively speaking A is least code change and least maintenance/operational overhead.
upvoted 9 times

JA2018 6 months, 2 weeks ago

from the same URL:

Is Amazon Aurora MySQL compatible?
Amazon Aurora is drop-in compatible with existing MySQL open-source databases and adds support for new releases regularly. This means you can easily migrate MySQL databases to and from Aurora using standard import/export tools or snapshots. It also means that most of the code, applications, drivers, and tools you already use with MySQL databases today can be used with Aurora with little or no change. This makes it easy to move applications between the two engines.

You can see the current Amazon Aurora MySQL release compatibility information in the documentation.
upvoted 1 times

pentium75 1 year, 5 months ago

Selected Answer: A

A: Minimal changes to the application code, < 1 second lag
B: Does not address the replication lag issue at all, requires code changes and adds overhead
C: Moving from managed RDS to self-managed database on EC2 is ADDING, not minimizing, overhead, PLUS it does not address the replication lag issue
D: DynamoDB is a NoSQL DB, would require MASSIVE changes to application code and probably even application logic
upvoted 5 times

Murtadhaceit 1 year, 6 months ago

Selected Answer: A

imho, B is not valid because it involves extra coding and the question specifically mentions no more coding. Therefore, replacing the current db with another one is not considered as more coding.
upvoted 3 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

Migrate the RDS MySQL database to Amazon Aurora MySQL

Migrate the RDS MySQL database to Amazon Aurora MySQL
Use Aurora Replicas for read scaling instead of RDS read replicas
Configure Aurora Auto Scaling to handle load spikes
Replace stored procedures with Aurora MySQL native functions
upvoted 2 times

MrAWSAssociate 1 year, 11 months ago

Selected Answer: A

First, Elasticache involves heavy change on application code. The question mentioned that "he solutions architect must minimize changes to the application code". Therefore B is not suitable and A is more appropriate for the question requirement.
upvoted 3 times

aaroncelestin 1 year, 9 months ago

... but migrating their ENTIRE prod database and its replicas to a new platform is not a heavy change?
upvoted 3 times

KMohsoe 2 years ago

Selected Answer: B

Why not B? Please explain to me.
upvoted 2 times

Terion 1 year, 8 months ago

It wouldn't have the most up to date info since it must no lag in relation to the main DB
upvoted 2 times

pentium75 1 year, 5 months ago

How would adding a cache "reduce the replication lag" between the primary instance and the read replicas? Plus, it would require "changes to the application code" that we want to avoid. The "AWS Lambda functions" would create "ongoing operational overhead" that we're also asked to avoid.
upvoted 2 times

kaushald 2 years, 3 months ago

Option A is the most appropriate solution for reducing replication lag without significant changes to the application code and minimizing ongoing operational overhead. Migrating the database to Amazon Aurora MySQL allows for improved replication performance and higher scalability compared to Amazon RDS for MySQL. Aurora Replicas provide faster replication, reducing the replication lag, and Aurora Auto Scaling ensures that there are enough Aurora Replicas to handle the incoming traffic. Additionally, Aurora MySQL native functions can replace the stored procedures, reducing the load on the database and improving performance.

Option B is not the best solution since adding an ElastiCache for Redis cluster does not address the replication lag issue, and the cache may not have the most up-to-date information. Additionally, replacing the stored procedures with AWS Lambda functions adds additional complexity and may not improve performance.
upvoted 5 times

njufi 1 year, 2 months ago

I agree with your explanation. Additionally, considering the requirement that "the read replicas must lag no more than 1 second behind the primary DB instance," it's crucial to ensure that Elasticache for Redis also maintains this tight synchronization window. This implies that the main RDS instance would need to synchronize an additional database, potentially exacerbating lag during peak times rather than alleviating it.
upvoted 1 times

[Removed] 2 years, 3 months ago

Selected Answer: B

a,b are confusing me..
i would like to go with b..
upvoted 1 times

bangfire 2 years, 3 months ago

Option B is incorrect because it suggests using ElastiCache for Redis as a caching layer in front of the database, but this would not necessarily reduce the replication lag on the read replicas. Additionally, it suggests replacing the stored procedures with

AWS Lambda functions, which may require significant changes to the application code.
upvoted 6 times

lizzard812 2 years, 2 months ago

Yes and moreover Redis requires app refactoring which is a solid operational overhead
upvoted 2 times

Nel8 2 years, 3 months ago

Selected Answer: B

By using ElastiCache you avoid a lot of common issues you might encounter. ElastiCache is a database caching solution. ElastiCache Redis per se, supports failover and Multi-AZ. And Most of all, ElastiCache is well suited to place in front of RDS.

Migrating a database such as option A, requires operational overhead.
upvoted 2 times

pentium75 1 year, 5 months ago

Database migration is one-time work, NOT "operational overhead". Plus, RDS for MySQL to Aurora with MySQL compatibility is not a big deal, and "minimizes changes to the application code" as requested.
upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: A

Aurora can have up to 15 read replicas - much faster than RDS
https://aws.amazon.com/rds/aurora/
upvoted 5 times

ChrisG1454 2 years, 3 months ago

" As a result, all Aurora Replicas return the same data for query results with minimal replica lag. This lag is usually much less than 100 milliseconds after the primary instance has written an update "

Reference:
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
upvoted 3 times

ChrisG1454 2 years, 3 months ago

You can invoke an Amazon Lambda function from an Amazon Aurora MySQL-Compatible Edition DB cluster with the "native function"....

https://docs.amazonaws.cn/en_us/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Integrating.Lambda.html
upvoted 1 times

jennyka76 2 years, 3 months ago

Answer - A
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_PostgreSQL.Replication.ReadReplicas.html
----
You can scale reads for your Amazon RDS for PostgreSQL DB instance by adding read replicas to the instance. As with other Amazon RDS database engines, RDS for PostgreSQL uses the native replication mechanisms of PostgreSQL to keep read replicas up to date with changes on the source DB. For general information about read replicas and Amazon RDS, see Working with read replicas.
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #338 *Topic 1*

A solutions architect must create a disaster recovery (DR) plan for a high-volume software as a service (SaaS) platform. All data for the platform is stored in an Amazon Aurora MySQL DB cluster.

The DR plan must replicate data to a secondary AWS Region.

Which solution will meet these requirements MOST cost-effectively?

A. Use MySQL binary log replication to an Aurora cluster in the secondary Region. Provision one DB instance for the Aurora cluster in the secondary Region.

B. Set up an Aurora global database for the DB cluster. When setup is complete, remove the DB instance from the secondary Region. **Most Voted**

C. Use AWS Database Migration Service (AWS DMS) to continuously replicate data to an Aurora cluster in the secondary Region. Remove the DB instance from the secondary Region.

D. Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region.

Correct Answer: *B*

Community vote distribution

B (58%) D (32%) Other (10%)

Comments

awsgeek75 Highly Voted 1 year, 5 months ago

Selected Answer: B

I originally went for D but now I think B is correct. D is active-active cluster so whereas B is active-passive (headless cluster) so it is cheaper than D.

https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/

upvoted 20 times

jennyka76 Highly Voted 2 years, 3 months ago

Answer - A
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Replication.CrossRegion.html
----

Before you begin
Before you can create an Aurora MySQL DB cluster that is a cross-Region read replica, you must turn on binary logging on your source Aurora MySQL DB cluster. Cross-region replication for Aurora MySQL uses MySQL binary replication to replay changes on the cross-Region read replica DB cluster.
upvoted 9 times

ChrisG1454 2 years, 3 months ago

The question states " The DR plan must replicate data to a "secondary" AWS Region."

In addition to Aurora Replicas, you have the following options for replication with Aurora MySQL:

Aurora MySQL DB clusters in different AWS Regions.

You can replicate data across multiple Regions by using an Aurora global database. For details, see High availability across AWS Regions with Aurora global databases

You can create an Aurora read replica of an Aurora MySQL DB cluster in a different AWS Region, by using MySQL binary log (binlog) replication. Each cluster can have up to five read replicas created this way, each in a different Region.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
upvoted 1 times

ChrisG1454 2 years, 3 months ago

The question is asking for the most cost-effective solution.
Aurora global databases are more expensive.

https://aws.amazon.com/rds/aurora/pricing/
upvoted 1 times

leoattf 2 years, 3 months ago

On this same URL you provided, there is a note highlighted, stating the following:
"Replication from the primary DB cluster to all secondaries is handled by the Aurora storage layer rather than by the database engine, so lag time for replicating changes is minimal—typically, less than 1 second. Keeping the database engine out of the replication process means that the database engine is dedicated to processing workloads. It also means that you don't need to configure or manage the Aurora MySQL binlog (binary logging) replication."

So, answer should be A
upvoted 2 times

leoattf 2 years, 3 months ago

Correction: So, answer should be D
upvoted 3 times

Yak_Yeti Most Recent 1 month, 1 week ago

Selected Answer: B

When you set up an Aurora Global Database, the storage layer (Aurora Storage) is continuously and automatically replicated across Regions — independent of whether you have DB instances (readers/writers) running in the secondary Region.
If you remove the DB instance from the secondary Region:
The replicated storage remains.
No compute (no DB instances) = No cost for compute, only storage costs.
BUT:
In a real failover event, you must promote the secondary cluster.
When you promote, you must first create a new DB instance in the secondary Region from the existing storage. This takes some time (typically a few minutes), because Aurora has to spin up the DB instance.
upvoted 1 times

tch 2 months, 4 weeks ago

Selected Answer: D

Answer could be D, as this is DR plan, answer B is ask to remove the DB instance which might not suitable for DR plan..... if you remove it... how to replicate data again? you setup again for next replication ?
upvoted 1 times

FlyingHawk 4 months, 2 weeks ago

Selected Answer: D

Giving the requirement is the DR plan for a high-volume software as a service (SaaS) platform, the RTO and RPO should be small, needs a quick recovery, D(warm standby) is a better choice than A (cold standby, pilot light)

upvoted 2 times

LeonSauveterre 6 months, 1 week ago

Selected Answer: D

A - No. Higher cost & higher latency.
B - No. A DB instance is mandatory in each Region for the global database to function.
C - Removing would make DR incomplete, because the secondary Region cannot process failovers or support read replicas.
D - Cost slightly higher, yes, but feasible & cost less than A.

Overall, I don't like this question ...

upvoted 2 times

theamachine 12 months ago

Selected Answer: B

Aurora Global Databases offer a cost-effective way to replicate data to a secondary region for disaster recovery. By removing the secondary DB instance after setup, you only pay for storage and minimal compute resources.

upvoted 2 times

thewalker 1 year, 4 months ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-global-database.html#aurora-global-database.advantages

upvoted 1 times

awsgeek75 1 year, 4 months ago

Wrong, while D will work, B is cheaper. This question is about DR, not cross region scaling

upvoted 2 times

upliftinghut 1 year, 5 months ago

Selected Answer: D

B is more cost-effective however because this is DR so when the region fails => still need a DB to fail over and if setting up a DB from snapshot at the time of failure will be risky => D is the answer

upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: B

"Achieve cost-effective multi-Region resiliency with Amazon Aurora Global Database headless clusters" is exactly the topic here. "A headless secondary Amazon Aurora database cluster is one without a database instance. This type of configuration can lower expenses for an Aurora global database."

https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/

upvoted 8 times

minagaboya 1 year, 6 months ago

shd be D i guess .. . Migrating the database to Amazon Aurora MySQL allows for improved replication performance and higher scalability compared to Amazon RDS for MySQL. Aurora Replicas provide faster replication, reducing the replication lag, and Aurora Auto Scaling ensures that there are enough Aurora Replicas to handle the incoming traffic. Additionally, Aurora MySQL native functions can replace the stored procedures, reducing the load on the database and improving performance.

Option B is not the best solution since adding an ElastiCache for Redis cluster does not address the replication lag issue, and the cache may not have the most up-to-date information. Additionally, replacing the stored procedures with AWS Lambda functions adds additional complexity and may not improve performance.

upvoted 2 times

pentium75 1 year, 5 months ago

This is about a different question

This is about a different question
upvoted 4 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: D

Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region
upvoted 1 times

vini15 1 year, 10 months ago

should be B for most cost effective solution.
see the link - Achieve cost-effective multi-Region resiliency with Amazon Aurora Global Database headless clusters
https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/
upvoted 2 times

luisgu 2 years ago

Selected Answer: B

MOST cost-effective --> B
See section "Creating a headless Aurora DB cluster in a secondary Region" on the link
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
"Although an Aurora global database requires at least one secondary Aurora DB cluster in a different AWS Region than the primary, you can use a headless configuration for the secondary cluster. A headless secondary Aurora DB cluster is one without a DB instance. This type of configuration can lower expenses for an Aurora global database. In an Aurora DB cluster, compute and storage are decoupled. Without the DB instance, you're not charged for compute, only for storage. If it's set up correctly, a headless secondary's storage volume is kept in-sync with the primary Aurora DB cluster."
upvoted 7 times

bsbs1234 1 year, 8 months ago

upvoted your message, but still think D is correct. Because the question is to design a DR plan.In case of DR, B need to create an instance in DR region manually.
upvoted 2 times

Abhineet9148232 2 years, 3 months ago

Selected Answer: D

D: With Amazon Aurora Global Database, you pay for replicated write I/Os between the primary Region and each secondary Region (in this case 1).

Not A because it achieves the same, would be equally costly and adds overhead.
upvoted 3 times

[Removed] 2 years, 3 months ago

Selected Answer: C

CCCCCC
upvoted 3 times

Steve_4542636 2 years, 3 months ago

Selected Answer: D

I think Amazon is looking for D here. I don' think A is intended because that would require knowledge of MySQL, which isn't what they are testing us on. Not option C because the question states large volume. If the volume were low, then DMS would be better. This question is not a good question.
upvoted 3 times

fkie4 2 years, 3 months ago

very true. Amazon wanna everyone to use AWS, why do they sell for MySQL?
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #339 *Topic 1*

A company has a custom application with embedded credentials that retrieves information from an Amazon RDS MySQL DB instance. Management says the application must be made more secure with the least amount of programming effort.

What should a solutions architect do to meet these requirements?

A. Use AWS Key Management Service (AWS KMS) to create keys. Configure the application to load the database credentials from AWS KMS. Enable automatic key rotation.

B. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Create an AWS Lambda function that rotates the credentials in Secret Manager.

C. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager. **Most Voted**

D. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Systems Manager Parameter Store. Configure the application to load the database credentials from Parameter Store. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Parameter Store.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

cloudbusting Highly Voted 2 years, 3 months ago

Parameter Store does not provide automatic credential rotation.

upvoted 16 times

Bhawesh Highly Voted 2 years, 3 months ago

Selected Answer: C

C. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager.

https://www.examtopics.com/discussions/amazon/view/46483-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 12 times

JA2018 Most Recent 6 months, 2 weeks ago

Selected Answer: C

Management says the application must be made more secure with the least amount of programming effort.

IMHO, this will rule out Lambda for this STEM.
upvoted 1 times

Gape4 11 months, 2 weeks ago

Selected Answer: C

credentials from Secrets Manager...
upvoted 2 times

d401c0d 1 year, 2 months ago

question is asking for "more secure with the least amount of programming effort." = Secrets Manager + Secretes Manager's built in rotation schedule instead of Lambda.
upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

A KMS is for encryption keys specifically so this is a long way of doing the credentials storage
B is too much work for rotation
C exactly what secrets manager is designed for
D You can do that if C wasn't an option
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

Store the RDS credentials in Secrets Manager
Configure the application to retrieve the credentials from Secrets Manager
Use Secrets Manager's built-in rotation to rotate the RDS credentials automatically
upvoted 2 times

Hades2231 1 year, 9 months ago

Selected Answer: C

Secrets Manager can handle the rotation, so no need for Lambda to rotate the keys.
upvoted 2 times

chen0305_099 1 year, 9 months ago

WHY NOT B ?
upvoted 1 times

StacyY 1 year, 9 months ago

B, we need lambda for password rotation, confirmed!
upvoted 2 times

Nikki013 1 year, 9 months ago

It is not needed for certain types RDS, including MySQL as Secrets Manager has built-in rotation capabilities for it:
https://aws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/
upvoted 3 times

Abrar2022 1 year, 12 months ago

Selected Answer: C

If you need your DB to store credentials then use AWS Secret Manager. System Manager Paramater Store is for CloudFormation (no rotation)
upvoted 2 times

AlessandraSAA 2 years, 3 months ago

why it's not A?
upvoted 4 times

MssP 2 years, 2 months ago

It is asking for credentials, not for encryption keys.
upvoted 7 times

PoisonBlack 2 years, 1 month ago

So credentials rotation is secrets manager and key rotation is KMS?
upvoted 3 times

bdp123 2 years, 3 months ago

Selected Answer: C

https://aws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/
upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: C

C is a valid solution for securing the custom application with the least amount of programming effort. It involves creating credentials on the RDS for MySQL database for the application user and storing them in AWS Secrets Manager. The application can then be configured to load the database credentials from Secrets Manager. Additionally, the solution includes setting up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager, which will automatically rotate the credentials at a specified interval without requiring any programming effort.
upvoted 4 times

bdp123 2 years, 3 months ago

Selected Answer: C

https://docs.aws.amazon.com/secretsmanager/latest/userguide/create_database_secret.html
upvoted 3 times

jennyka76 2 years, 3 months ago

Answer - C
https://ws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #340 *Topic 1*

A media company hosts its website on AWS. The website application's architecture includes a fleet of Amazon EC2 instances behind an Application Load Balancer (ALB) and a database that is hosted on Amazon Aurora. The company's cybersecurity team reports that the application is vulnerable to SQL injection.

How should the company resolve this issue?

A. Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF. **Most Voted**

B. Create an ALB listener rule to reply to SQL injections with a fixed response.

C. Subscribe to AWS Shield Advanced to block all SQL injection attempts automatically.

D. Set up Amazon Inspector to block all SQL injection attempts automatically.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

Bhawesh Highly Voted 1 year, 3 months ago

Selected Answer: A

A. Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF.

SQL Injection - AWS WAF
DDoS - AWS Shield

upvoted 24 times

jennyka76 Highly Voted 1 year, 3 months ago

Answer - A
https://aws.amazon.com/premiumsupport/knowledge-center/waf-block-common-attacks/#:~:text=To%20protect%20your%20applications%20against,%2C%20query%20string%2C%20or%20URI.

Protect against SQL injection and cross-site scripting
To protect your applications against SQL injection and cross-site scripting (XSS) attacks, use the built-in SQL injection and cross-site scripting engines. Remember that attacks can be performed on different parts of the HTTP request, such as the HTTP header, query string, or URI. Configure the AWS WAF rules to inspect different parts of the HTTP request against the built-in mitigation engines.

upvoted 8 times

wsdasdasdqwdaw Most Recent 7 months, 3 weeks ago

AWS WAF - for SQL Injection ---> A

AWS Shield - for DDOS
Amazon Inspector - for automated security assessment, like known vulnerability
upvoted 4 times

Guru4Cloud 9 months ago

Selected Answer: A

° Use AWS WAF in front of the Application Load Balancer
° Configure appropriate WAF web ACLs to detect and block SQL injection patterns
The key points:
° Website hosted on EC2 behind an ALB with Aurora database
° Application is vulnerable to SQL injection attacks
° AWS WAF is designed to detect and block SQL injection and other common web exploits. It can be placed in front of the ALB to inspect all incoming requests. WAF rules can identify malicious SQL patterns and block them.
upvoted 2 times

KMohsoe 1 year ago

Selected Answer: A

SQL injection -> WAF
upvoted 2 times

lexotan 1 year, 1 month ago

Selected Answer: A

WAF is the right one
upvoted 2 times

akram_akram 1 year, 2 months ago

Selected Answer: A

SQL Injection - AWS WAF
DDoS - AWS Shield
upvoted 2 times

movva12 1 year, 2 months ago

Answer C - Shield Advanced (WAF + Firewall Manager)
upvoted 1 times

fkie4 1 year, 3 months ago

Selected Answer: A

It is A. I am happy to see Amazon gives out score like this...
upvoted 2 times

LuckyAro 1 year, 3 months ago

Selected Answer: A

AWS WAF is a managed service that protects web applications from common web exploits that could affect application availability, compromise security, or consume excessive resources. AWS WAF enables customers to create custom rules that block common attack patterns, such as SQL injection attacks.

By using AWS WAF in front of the ALB and associating the appropriate web ACLs with AWS WAF, the company can protect its website application from SQL injection attacks. AWS WAF will inspect incoming traffic to the website application and block requests that match the defined SQL injection patterns in the web ACLs. This will help to prevent SQL injection attacks from reaching the application, thereby improving the overall security posture of the application.
upvoted 3 times

LuckyAro 1 year, 3 months ago

B, C, and D are not the best solutions for this issue. Replying to SQL injections with a fixed response
(B) is not a recommended approach as it does not actually fix the vulnerability, but only masks the issue. Subscribing to AWS

Shield Advanced
(C) is useful to protect against DDoS attacks but does not protect against SQL injection vulnerabilities. Amazon Inspector
(D) is a vulnerability assessment tool and can identify vulnerabilities but cannot block attacks in real-time.
upvoted 3 times

pbpally 1 year, 3 months ago

Selected Answer: A

Bhawesh answers it perfect so I'm avoiding redundancy but agree on it being A.
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #341 *Topic 1*

A company has an Amazon S3 data lake that is governed by AWS Lake Formation. The company wants to create a visualization in Amazon QuickSight by joining the data in the data lake with operational data that is stored in an Amazon Aurora MySQL database. The company wants to enforce column-level authorization so that the company's marketing team can access only a subset of columns in the database.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon EMR to ingest the data directly from the database to the QuickSight SPICE engine. Include only the required columns.

B. Use AWS Glue Studio to ingest the data from the database to the S3 data lake. Attach an IAM policy to the QuickSight users to enforce column-level access control. Use Amazon S3 as the data source in QuickSight.

C. Use AWS Glue Elastic Views to create a materialized view for the database in Amazon S3. Create an S3 bucket policy to enforce column-level access control for the QuickSight users. Use Amazon S3 as the data source in QuickSight.

D. Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight. Most Voted

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

K0nAn Highly Voted 1 year, 9 months ago

Selected Answer: D

This solution leverages AWS Lake Formation to ingest data from the Aurora MySQL database into the S3 data lake, while enforcing column-level access control for QuickSight users. Lake Formation can be used to create and manage the data lake's metadata and enforce security and governance policies, including column-level access control. This solution then uses Amazon Athena as the data source in QuickSight to query the data in the S3 data lake. This solution minimizes operational overhead by leveraging AWS services to manage and secure the data, and by using a standard query service (Amazon Athena) to provide a SQL interface to the data.

upvoted 14 times

jennyka76 Highly Voted 1 year, 9 months ago

Answer - D
https://aws.amazon.com/blogs/big-data/enforce-column-level-authorization-with-amazon-quicksight-and-aws-lake-formation/
upvoted 10 times

awsgeek75 Most Recent 10 months, 3 weeks ago

Selected Answer: D

https://docs.aws.amazon.com/lake-formation/latest/dg/workflows-about.html
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: D

Use a Lake Formation blueprint to ingest data from the Aurora database into the S3 data lake
Leverage Lake Formation to enforce column-level access control for the marketing team
Use Amazon Athena as the data source in QuickSight
The key points:

Need to join S3 data lake data with Aurora MySQL data
Require column-level access controls for marketing team in QuickSight
Minimize operational overhead
upvoted 4 times

LuckyAro 1 year, 9 months ago

Selected Answer: D

Using a Lake Formation blueprint to ingest the data from the database to the S3 data lake, using Lake Formation to enforce column-level access control for the QuickSight users, and using Amazon Athena as the data source in QuickSight. This solution requires the least operational overhead as it utilizes the features provided by AWS Lake Formation to enforce column-level authorization, which simplifies the process and reduces the need for additional configuration and maintenance.
upvoted 5 times

Bhawesh 1 year, 9 months ago

Selected Answer: D

D. Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight.

https://www.examtopics.com/discussions/amazon/view/80865-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #342 *Topic 1*

A transaction processing company has weekly scripted batch jobs that run on Amazon EC2 instances. The EC2 instances are in an Auto Scaling group. The number of transactions can vary, but the baseline CPU utilization that is noted on each run is at least 60%. The company needs to provision the capacity 30 minutes before the jobs run.

Currently, engineers complete this task by manually modifying the Auto Scaling group parameters. The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. The company needs an automated way to modify the Auto Scaling group's desired capacity.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create a dynamic scaling policy for the Auto Scaling group. Configure the policy to scale based on the CPU utilization metric. Set the target value for the metric to 60%.

B. Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.

C. Create a predictive scaling policy for the Auto Scaling group. Configure the policy to scale based on forecast. Set the scaling metric to CPU utilization. Set the target value for the metric to 60%. In the policy, set the instances to pre-launch 30 minutes before the jobs run. **Most Voted**

D. Create an Amazon EventBridge event to invoke an AWS Lambda function when the CPU utilization metric value for the Auto Scaling group reaches 60%. Configure the Lambda function to increase the Auto Scaling group's desired capacity and maximum capacity by 20%.

Correct Answer: *C*

Community vote distribution

C (68%) B (29%) A

Comments

fkie4 Highly Voted 2 years, 2 months ago

Selected Answer: C

B is NOT correct. the question said "The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts.".
answer B said "Set the appropriate desired capacity, minimum capacity, and maximum capacity".

how can someone set desired capacity if he has no resources to analyze the required capacity.
Read carefully Amigo
upvoted 26 times

omoakin 2 years ago

scheduled scaling....
upvoted 4 times

jjcode 1 year, 3 months ago

works loads can vary, how can you predict something that is random?
upvoted 2 times

ealpuche 2 years ago

But you can make a vague estimation according to the resources used; you don't need to make machine learning models to do that. You only need common sense.
upvoted 1 times

Murtadhaceit 1 year, 6 months ago

Your explanation is contradicting your answer. Since "the company does not have the resources to analyze the required capacity trend for the ASG", how come they can create and ASG based on a historic trend?
C doesn't make sense for me.
upvoted 4 times

neverdie Highly Voted 2 years, 2 months ago

Selected Answer: B

A scheduled scaling policy allows you to set up specific times for your Auto Scaling group to scale out or scale in. By creating a scheduled scaling policy for the Auto Scaling group, you can set the appropriate desired capacity, minimum capacity, and maximum capacity, and set the recurrence to weekly. You can then set the start time to 30 minutes before the batch jobs run, ensuring that the required capacity is provisioned before the jobs run.

Option C, creating a predictive scaling policy for the Auto Scaling group, is not necessary in this scenario since the company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. This would require analyzing the required capacity trends for the Auto Scaling group counts to determine the appropriate scaling policy.
upvoted 6 times

[Removed] 2 years, 2 months ago

(typo above) C is correct..
upvoted 1 times

MssP 2 years, 2 months ago

Look at fkie4 comment... no way to know desired capacity!!! -> B not correct
upvoted 1 times

Lalo 2 years ago

the text says
1.-"A transaction processing company has weekly scripted batch jobs", there is a Schedule
2.-" The company does not have the resources to analyze the required capacity trends for the Auto Scaling " Do not use
the answer is B
upvoted 4 times

[Removed] 2 years, 2 months ago

B is correct. "Predictive scaling uses machine learning to predict capacity requirements based on historical data from CloudWatch.", meaning the company does not have to analyze the capacity trends themselves.
https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-predictive-scaling.html
upvoted 3 times

bora4motion Most Recent 1 month ago

Selected Answer: B

C is a bit weird - it says "predictive"while the question says "there's no budget to analyze".

upvoted 1 times

bora4motion 1 month, 1 week ago

Selected Answer: B

B is correct
You know the schedule: weekly jobs → exact match.
You don't need to analyze utilization trends → simple config.
1-time setup → minimal overhead.
C is wrong, at first glance, looks great.
 BUT it requires historical data and analysis — which the company cannot support.
Setup and tuning introduce overhead.
 Breaks the "company lacks resources to analyze trends" condition.
 Incorrect.

upvoted 1 times

9820ad3 2 months, 1 week ago

Selected Answer: B

B is correct answer ....

upvoted 1 times

Denise123 5 months, 2 weeks ago

Selected Answer: C

Option A - Dynamic scaling policy is reactive not proactive.
Option B - Scheduled policy is not suitable as it says the work loads can vary
Option C - Correct
Option D - Irrelevant

upvoted 1 times

EllenLiu 5 months, 3 weeks ago

Selected Answer: C

"The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. " ==> this means this company can take advantage of cloudwatch metric and auto scaling on AWS to set up a predictive scaling group, so that no need to have any resources for this kind of task any more.??

upvoted 1 times

studydue 7 months, 1 week ago

Answer should be B
Pref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-predictive-scaling.html
Pref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/predictive-scaling-policy-overview.html
keyword for this question: "s. The company does not have
the resources to analyze the required capacity trends for the Auto Scaling group counts"

upvoted 2 times

Hkayne 1 year, 1 month ago

Selected Answer: C

B or C.
I think C because the company needs an automated way to modify the autoscaling desired capacity

upvoted 1 times

jjcode 1 year, 3 months ago

How does C works with : transactions can vary, clearly C is designed for workloads that are predictable, if the transactions can vary then predictive scaling will not work. The only one that will work is scheduled since its based on time not workload intensity.

upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: C

C per https://docs.aws.amazon.com/autoscaling/ec2/userguide/predictive-scaling-create-policy.html.

B is out because it wants the company to 'set the desired/minimum/maximum capacity' but "the company does not have the resources to analyze the required capacity".

upvoted 6 times

Cyberkayu 1 year, 5 months ago

Lambda did not appear to take over scripting/batch job, what a surprise

upvoted 4 times

daniel1 1 year, 7 months ago

Selected Answer: B

From GPT4:
mong the provided options, creating a scheduled scaling policy (Option B) is the most direct and efficient way to ensure that the necessary capacity is provisioned 30 minutes before the weekly batch jobs run, with the least operational overhead. Here's a breakdown of Option B:

B. Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.

Scheduled scaling allows you to change the desired capacity of your Auto Scaling group based on a schedule. In this case, setting the recurrence to weekly and adjusting the start time to 30 minutes before the batch jobs run will ensure that the necessary capacity is available when needed, without requiring manual intervention.

upvoted 5 times

TheFivePips 1 year, 3 months ago

yeah chatgpt told me this, so maybe dont take its word as gospel:

Upon reviewing the question again, it appears that the requirements emphasize the need to provision capacity 30 minutes before the batch jobs run and the company's constraint of not having resources to analyze capacity trends. In this context, the most suitable solution is C.

Predictive Scaling can use historical data to forecast future capacity needs.
Configuring the policy to scale based on CPU utilization with a target value of 60% aligns with the baseline CPU utilization mentioned in the scenario.
Setting instances to pre-launch 30 minutes before the jobs run provides the desired capacity just in time.

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: C

Predictive scaling: increases the number of EC2 instances in your Auto Scaling group in advance of daily and weekly patterns in traffic flows. If you have regular patterns of traffic increases use predictive scaling, to help you scale faster by launching capacity in advance of forecasted load. You don't have to spend time reviewing your application's load patterns and trying to schedule the right amount of capacity using scheduled scaling. Predictive scaling uses machine learning to predict capacity requirements based on historical data from CloudWatch. The machine learning algorithm consumes the available historical data and calculates capacity that best fits the historical load pattern, and then continuously learns based on new data to make future forecasts more accurate.

upvoted 2 times

bsbs1234 1 year, 8 months ago

should be C. Question does not say how long the job will run. don't know when to set the end time in the schedule policy.

upvoted 2 times

MrAWSAssociate 1 year, 11 months ago

Selected Answer: C

C is correct!

upvoted 2 times

Abrar2022 1 year, 12 months ago

Selected Answer: C

if the baseline CPU utilization is 60%, then that's enough information needed to determaine you to predict some aspect of the usage in the future. So key word "predictive" judging by past usage.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #343 *Topic 1*

A solutions architect is designing a company's disaster recovery (DR) architecture. The company has a MySQL database that runs on an Amazon EC2 instance in a private subnet with scheduled backup. The DR design needs to include multiple AWS Regions.

Which solution will meet these requirements with the LEAST operational overhead?

A. Migrate the MySQL database to multiple EC2 instances. Configure a standby EC2 instance in the DR Region. Turn on replication.

B. Migrate the MySQL database to Amazon RDS. Use a Multi-AZ deployment. Turn on read replication for the primary DB instance in the different Availability Zones.

C. Migrate the MySQL database to an Amazon Aurora global database. Host the primary DB cluster in the primary Region. Host the secondary DB cluster in the DR Region. **Most Voted**

D. Store the scheduled backup of the MySQL database in an Amazon S3 bucket that is configured for S3 Cross-Region Replication (CRR). Use the data backup to restore the database in the DR Region.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

LuckyAro Highly Voted 1 year, 9 months ago

C: Migrate MySQL database to an Amazon Aurora global database is the best solution because it requires minimal operational overhead. Aurora is a managed service that provides automatic failover, so standby instances do not need to be manually configured. The primary DB cluster can be hosted in the primary Region, and the secondary DB cluster can be hosted in the DR Region. This approach ensures that the data is always available and up-to-date in multiple Regions, without requiring significant manual intervention.

upvoted 10 times

AlessandraSAA Highly Voted 1 year, 9 months ago

Selected Answer: C

A. Multiple EC2 instances to be configured and updated manually in case of DR.
B. Amazon RDS=Multi-AZ while it asks to be multi-region
C. correct, see comment from LuckyAro

C. correct, see comment from LuckyAro
D. Manual process to start the DR, therefore same limitation as answer A
upvoted 10 times

gulmichamagaun5 Most Recent 11 months, 2 weeks ago

hello friends, question required: The DR design needs to include multiple AWS Regions, but the correct answer is B, how it comes, because the DR here is on AZ not Different Region so the i would go with D
upvoted 1 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: C

LEAST operational overhead = Serverless = Amazon Aurora global database
upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: C

Amazon Aurora global database can span and replicate DB Servers between multiple AWS Regions. And also compatible with MySQL.
upvoted 2 times

GalileoEC2 1 year, 8 months ago

C, Why B? B is multi zone in one region, C is multi region as it was requested
upvoted 2 times

lucdt4 1 year, 6 months ago

" The DR design needs to include multiple AWS Regions."
with the requirement "DR SITE multiple AWS region" -> B is wrong, because it deploy multy AZ (this is not multi region)
upvoted 2 times

KZM 1 year, 9 months ago

Amazon Aurora global database can span and replicate DB Servers between multiple AWS Regions. And also compatible with MySQL.
upvoted 4 times

LuckyAro 1 year, 9 months ago

With dynamic scaling, the Auto Scaling group will automatically adjust the number of instances based on the actual workload. The target value for the CPU utilization metric is set to 60%, which is the baseline CPU utilization that is noted on each run, indicating that this is a reasonable level of utilization for the workload. This solution does not require any scheduling or forecasting, reducing the operational overhead.
upvoted 1 times

LuckyAro 1 year, 9 months ago

Sorry, Posted right answer to the wrong question, mistakenly clicked the next question, sorry.
upvoted 6 times

geekgirl22 1 year, 9 months ago

C is the answer as RDS is only multi-zone not multi region.
upvoted 2 times

bdp123 1 year, 9 months ago

Selected Answer: C

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
upvoted 2 times

SMAZ 1 year, 9 months ago

C
option A has operation overhead whereas option C not.
upvoted 2 times

alexman 1 year, 9 months ago

Selected Answer: C

C mentions multiple regions. Option B is within the same region

upvoted 4 times

jennyka76 1 year, 9 months ago

ANSWER - B ?? NOT SURE

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #344 *Topic 1*

A company has a Java application that uses Amazon Simple Queue Service (Amazon SQS) to parse messages. The application cannot parse messages that are larger than 256 KB in size. The company wants to implement a solution to give the application the ability to parse messages as large as 50 MB.

Which solution will meet these requirements with the FEWEST changes to the code?

A. Use the Amazon SQS Extended Client Library for Java to host messages that are larger than 256 KB in Amazon S3. **Most Voted**

B. Use Amazon EventBridge to post large messages from the application instead of Amazon SQS.

C. Change the limit in Amazon SQS to handle messages that are larger than 256 KB.

D. Store messages that are larger than 256 KB in Amazon Elastic File System (Amazon EFS). Configure Amazon SQS to reference this location in the messages.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

LuckyAro Highly Voted 2 years, 3 months ago

Selected Answer: A

A. Use the Amazon SQS Extended Client Library for Java to host messages that are larger than 256 KB in Amazon S3.

Amazon SQS has a limit of 256 KB for the size of messages. To handle messages larger than 256 KB, the Amazon SQS Extended Client Library for Java can be used. This library allows messages larger than 256 KB to be stored in Amazon S3 and provides a way to retrieve and process them. Using this solution, the application code can remain largely unchanged while still being able to process messages up to 50 MB in size.

upvoted 17 times

Neha999 Highly Voted 2 years, 3 months ago

A
For messages > 256 KB, use Amazon SQS Extended Client Library for Java
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/quotas-messages.html

upvoted 7 times

tonywang0418 Most Recent 7 months, 3 weeks ago

who would know this...

upvoted 3 times

Gape4 11 months, 2 weeks ago

Selected Answer: A

To send messages larger than 256 KiB, you can use the Amazon SQS Extended Client Library for Java...

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: A

The Amazon SQS Extended Client Library for Java enables you to manage Amazon SQS message payloads with Amazon S3. This is especially useful for storing and retrieving messages with a message payload size greater than the current SQS limit of 256 KB, up to a maximum of 2 GB.

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

The SQS Extended Client Library enables storing large payloads in S3 while referenced via SQS. The application code can stay almost entirely unchanged - it sends/receives SQS messages normally. The library handles transparently routing the large payloads to S3 behind the scenes

upvoted 2 times

james2033 1 year, 10 months ago

Selected Answer: A

Quote "The Amazon SQS Extended Client Library for Java enables you to manage Amazon SQS message payloads with Amazon S3." and "An extension to the Amazon SQS client that enables sending and receiving messages up to 2GB via Amazon S3." at https://github.com/awslabs/amazon-sqs-java-extended-client-lib

upvoted 2 times

Abrar2022 1 year, 12 months ago

Selected Answer: A

Amazon SQS has a limit of 256 KB for the size of messages.

To handle messages larger than 256 KB, the Amazon SQS Extended Client Library for Java can be used.

upvoted 2 times

gold4otas 2 years, 2 months ago

The Amazon SQS Extended Client Library for Java enables you to publish messages that are greater than the current SQS limit of 256 KB, up to a maximum of 2 GB.

https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-s3-messages.html

upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: A

https://github.com/awslabs/amazon-sqs-java-extended-client-lib

upvoted 4 times

Arathore 2 years, 3 months ago

Selected Answer: A

To send messages larger than 256 KiB, you can use the Amazon SQS Extended Client Library for Java. This library allows you to send an Amazon SQS message that contains a reference to a message payload in Amazon S3. The maximum payload size is 2 GB.

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #345 *Topic 1*

A company wants to restrict access to the content of one of its main web applications and to protect the content by using authorization techniques available on AWS. The company wants to implement a serverless architecture and an authentication solution for fewer than 100 users. The solution needs to integrate with the main web application and serve web content globally. The solution must also scale as the company's user base grows while providing the lowest login latency possible.

Which solution will meet these requirements MOST cost-effectively?

A. Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally. **Most Voted**

B. Use AWS Directory Service for Microsoft Active Directory for authentication. Use AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally.

C. Use Amazon Cognito for authentication. Use AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally.

D. Use AWS Directory Service for Microsoft Active Directory for authentication. Use Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

Lonojack Highly Voted 1 year, 9 months ago

Selected Answer: A

CloudFront=globally
Lambda@edge = Authorization/ Latency
Cognito=Authentication for Web apps

upvoted 15 times

Lin878 Most Recent 6 months ago

Selected Answer: A

https://aws.amazon.com/blogs/networking-and-content-delivery/external-server-authorization-with-lambdaedge/

upvoted 2 times

upvoted 2 times

Cyberkayu 11 months, 4 weeks ago

fewer than 100 users but scattered around the globe, lowest latency.

Should have do nothing, most cost effective.

upvoted 3 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: A

Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally

upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: A

Amazon Cognito is a serverless authentication service that can be used to easily add user sign-up and authentication to web and mobile apps. It is a good choice for this scenario because it is scalable and can handle a small number of users without any additional costs.

Lambda@Edge is a serverless compute service that can be used to run code at the edge of the AWS network. It is a good choice for this scenario because it can be used to perform authorization checks at the edge, which can improve the login latency.

Amazon CloudFront is a content delivery network (CDN) that can be used to serve web content globally. It is a good choice for this scenario because it can cache web content closer to users, which can improve the performance of the web application.

upvoted 4 times

antropaws 1 year, 6 months ago

Selected Answer: A

A is perfect.

upvoted 2 times

kraken21 1 year, 8 months ago

Selected Answer: A

Lambda@Edge for authorization
https://aws.amazon.com/blogs/networking-and-content-delivery/adding-http-security-headers-using-lambdaedge-and-amazon-cloudfront/

upvoted 3 times

LuckyAro 1 year, 9 months ago

Selected Answer: A

Amazon CloudFront is a global content delivery network (CDN) service that can securely deliver web content, videos, and APIs at scale. It integrates with Cognito for authentication and with Lambda@Edge for authorization, making it an ideal choice for serving web content globally.

Lambda@Edge is a service that lets you run AWS Lambda functions globally closer to users, providing lower latency and faster response times. It can also handle authorization logic at the edge to secure content in CloudFront. For this scenario, Lambda@Edge can provide authorization for the web application while leveraging the low-latency benefit of running at the edge.

upvoted 3 times

bdp123 1 year, 9 months ago

Selected Answer: A

CloudFront to serve globally

upvoted 2 times

SMAZ 1 year, 9 months ago

A
Amazon Cognito for authentication and Lambda@Edge for authorizatioN, Amazon CloudFront to serve the web application globally provides low-latency content delivery

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #346 *Topic 1*

A company has an aging network-attached storage (NAS) array in its data center. The NAS array presents SMB shares and NFS shares to client workstations. The company does not want to purchase a new NAS array. The company also does not want to incur the cost of renewing the NAS array's support contract. Some of the data is accessed frequently, but much of the data is inactive.

A solutions architect needs to implement a solution that migrates the data to Amazon S3, uses S3 Lifecycle policies, and maintains the same look and feel for the client workstations. The solutions architect has identified AWS Storage Gateway as part of the solution.

Which type of storage gateway should the solutions architect provision to meet these requirements?

A. Volume Gateway

B. Tape Gateway

C. Amazon FSx File Gateway

D. Amazon S3 File Gateway **Most Voted**

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

LuckyAro Highly Voted 1 year, 9 months ago

Selected Answer: D

Amazon S3 File Gateway provides on-premises applications with access to virtually unlimited cloud storage using NFS and SMB file interfaces. It seamlessly moves frequently accessed data to a low-latency cache while storing colder data in Amazon S3, using S3 Lifecycle policies to transition data between storage classes over time.

In this case, the company's aging NAS array can be replaced with an Amazon S3 File Gateway that presents the same NFS and SMB shares to the client workstations. The data can then be migrated to Amazon S3 and managed using S3 Lifecycle policies

upvoted 17 times

everfly Highly Voted 1 year, 9 months ago

Selected Answer: D

Amazon S3 File Gateway provides a file interface to objects stored in S3. It can be used for a file-based interface with S3, which allows the company to migrate their NAS array data to S3 while maintaining the same look and feel for client workstations. Amazon S3 File Gateway supports SMB and NFS protocols, which will allow clients to continue to access the data using these protocols. Additionally, Amazon S3 Lifecycle policies can be used to automate the movement of data to lower-cost storage tiers, reducing the storage cost of inactive data.

upvoted 7 times

pentium75 Most Recent 11 months, 2 weeks ago

Selected Answer: D

A - provides virtual disk via iSCSI
B - provides virtual tape via iSCSI
C - provides access to FSx via SMB

upvoted 5 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: D

The Amazon S3 File Gateway enables you to store and retrieve objects in Amazon Simple Storage Service (S3) using file protocols such as Network File System (NFS) and Server Message Block (SMB).

upvoted 4 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: D

It provides an easy way to lift-and-shift file data from the existing NAS to Amazon S3. The S3 File Gateway presents SMB and NFS file shares that client workstations can access just like the NAS shares.
Behind the scenes, it moves the file data to S3 storage, storing it durably and cost-effectively.
S3 Lifecycle policies can be used to transition less frequently accessed data to lower-cost S3 storage tiers like S3 Glacier.
From the client workstation perspective, access to files feels seamless and unchanged after migration to S3. The S3 File Gateway handles the underlying data transfers.
It is a simple, low-cost gateway option tailored for basic file share migration use cases.

upvoted 4 times

james2033 1 year, 4 months ago

Selected Answer: D

- Volume Gateway: https://aws.amazon.com/storagegateway/volume/ (Remove A, related iSCSI)

- Tape Gateway https://aws.amazon.com/storagegateway/vtl/ (Remove B)

- Amazon FSx File Gateway https://aws.amazon.com/storagegateway/file/fsx/ (C)

- Why not choose C? Because need working with Amazon S3. (Answer D, and it is correct answer)
https://aws.amazon.com/storagegateway/file/s3/

upvoted 4 times

siyam008 1 year, 9 months ago

Selected Answer: D

https://aws.amazon.com/blogs/storage/how-to-create-smb-file-shares-with-aws-storage-gateway-using-hyper-v/

upvoted 3 times

bdp123 1 year, 9 months ago

Selected Answer: D

https://aws.amazon.com/about-aws/whats-new/2018/06/aws-storage-gateway-adds-smb-support-to-store-objects-in-amazon-s3/

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #347 *Topic 1*

A company has an application that is running on Amazon EC2 instances. A solutions architect has standardized the company on a particular instance family and various instance sizes based on the current needs of the company.

The company wants to maximize cost savings for the application over the next 3 years. The company needs to be able to change the instance family and sizes in the next 6 months based on application popularity and usage.

Which solution will meet these requirements MOST cost-effectively?

A. Compute Savings Plan **Most Voted**

B. EC2 Instance Savings Plan

C. Zonal Reserved Instances

D. Standard Reserved Instances

Correct Answer: *A*

Community vote distribution

A (83%) Other (17%)

Comments

AlmeroSenior Highly Voted 1 year, 9 months ago

Selected Answer: A

Read Carefully guys , They need to be able to change FAMILY , and although EC2 Savings has a higher discount , its clearly documented as not allowed >

EC2 Instance Savings Plans provide savings up to 72 percent off On-Demand, in exchange for a commitment to a specific instance family in a chosen AWS Region (for example, M5 in Virginia). These plans automatically apply to usage regardless of size (for example, m5.xlarge, m5.2xlarge, etc.), OS (for example, Windows, Linux, etc.), and tenancy (Host, Dedicated, Default) within the specified family in a Region.

upvoted 22 times

FFO 1 year, 7 months ago

Savings Plans are a flexible pricing model that offer low prices on Amazon EC2, AWS Lambda, and AWS Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in $/hour) for a 1 or 3 year term. When you sign up for a Savings Plan, you will be charged the discounted Savings Plans price for your usage up to your commitment.

The company wants savings over the next 3 years but wants to change the instance type in 6 months. This invalidates A
upvoted 5 times

FFO 1 year, 7 months ago

Disregard! found more information:
We recommend Savings Plans (over Reserved Instances). Like Reserved Instances, Savings Plans offer lower prices (up to 72% savings compared to On-Demand Instance pricing). In addition, Savings Plans offer you the flexibility to change your usage as your needs evolve. For example, with Compute Savings Plans, lower prices will automatically apply when you change from C4 to C6g instances, shift a workload from EU (Ireland) to EU (London), or move a workload from Amazon EC2 to AWS Fargate or AWS Lambda.
https://aws.amazon.com/ec2/pricing/reserved-instances/pricing/
upvoted 3 times

awsgeek75 Highly Voted 10 months, 3 weeks ago

Selected Answer: A

https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/savings-plans.html

Compute Savings Plans provide the most flexibility and help to reduce your costs by up to 66% (just like Convertible RIs). These plans automatically apply to EC2 instance usage regardless of instance family...

EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% (just like Standard RIs) in exchange for commitment to usage of individual instance families

Instance Savings "locks" you in that instance family which is not desired by the company hence A is the best plan as they can change the instance family anytime
upvoted 10 times

awsgeek75 10 months, 3 weeks ago

Also, don't forget, the minimum commitment for both of these plans is 1 year and the company wants the ability to change in 6 months so it has to be a plan which allows changing of instance within the commitment window (no refunds!)
upvoted 4 times

xBUGx Most Recent 8 months, 1 week ago

Selected Answer: A

EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% in exchange for commitment to usage of individual instance families in a region (e.g. M5 usage in N. Virginia). This automatically reduces your cost on the selected instance family in that region regardless of AZ, size, OS or tenancy. ***EC2 Instance Savings Plans give you the flexibility to change your usage between instances within a family in that region.*** For example, you can move from c5.xlarge running Windows to c5.2xlarge running Linux and automatically benefit from the Savings Plans prices.
https://aws.amazon.com/savingsplans/faq/#:~:text=EC2%20Instance%20Savings%20Plans%20give,from%20the%20Savings%20Plans%20prices.
upvoted 4 times

Mican07 11 months ago

B is the definite answer
upvoted 1 times

pentium75 11 months, 2 weeks ago

Selected Answer: A

B does not allow changing the instance family, despite all the ChatGPT-based answers claiming the opposite
upvoted 3 times

meowruki 1 year ago

Selected Answer: A

While EC2 Instance Savings Plans also provide cost savings over On-Demand pricing, they offer less flexibility in terms of changing instance families. They provide a discount in excha
upvoted 2 times

hungta 1 year ago

Selected Answer: B

EC2 Instance Savings Plans is most saving. And it is enough for required flexibility

EC2 Instance Savings Plans is most saving. And it is enough for required flexibility
EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% (just like Standard RIs) in exchange for commitment to usage of individual instance families in a Region (for example, M5 usage in N. Virginia). This automatically reduces your cost on the selected instance family in that region regardless of AZ, size, operating system, or tenancy. EC2 Instance Savings Plans give you the flexibility to change your usage between instances within a family in that Region. For example, you can move from c5.xlarge running Windows to c5.2xlarge running Linux and automatically benefit from the Savings Plans prices.
upvoted 1 times

pentium75 11 months, 2 weeks ago

But it does not allow changing the instance family, which is a requirement here.
upvoted 3 times

Jazz888 6 months, 1 week ago

You voted against yourself. Did you mean to vote A?
upvoted 1 times

dilaaziz 1 year ago

Selected Answer: A

https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/savings-plans.html
upvoted 2 times

EdenWang 1 year, 1 month ago

Selected Answer: B

The most cost-effective solution that meets the company's requirements would be B. EC2 Instance Savings Plan.

EC2 Instance Savings Plans provide significant cost savings, allowing the company to commit to a consistent amount of usage (measured in $/hour) for a 1- or 3-year term, and in return, receive a discount on the hourly rate for the instances that match the attributes of the plan.

With EC2 Instance Savings Plans, the company can benefit from the flexibility to change the instance family and sizes over the next 3 years, which aligns with their requirement to adjust based on application popularity and usage.

This option provides the best balance of cost savings and flexibility, making it the most suitable choice for the company's needs.
upvoted 2 times

pentium75 11 months, 2 weeks ago

"With EC2 Instance Savings Plans, the company can benefit from the flexibility to change the instance family" NO, this is simply wrong. Is this from ChatGPT?

"EC2 Instance Savings Plans provide savings up to 72 percent off On-Demand, in exchange for a commitment to a specific instance family (!) in a chosen AWS Region ... With an EC2 Instance Savings Plan, you can change your instance size within the instance family (!)".

https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 4 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: A

Change instance family = Compute Savings Plans
upvoted 4 times

Wayne23Fang 1 year, 3 months ago

Selected Answer: A

D is not right. D. Standard Reserved Instances. should be Convertible Reserved Instances if you need additional flexibility, such as the ability to use different instance families, operating systems.
upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: B

The key factors are:

The key factors are:

Need to maximize cost savings over 3 years
Ability to change instance family and sizes in 6 months
Standardized on a particular instance family for now
upvoted 2 times

awsgeek75 10 months, 3 weeks ago

"Ability to change instance family and sizes in 6 months" is not allowed in Instance Savings plan so B is wrong
upvoted 2 times

Kiki_Pass 1 year, 4 months ago

Why not C? Can do with Convertible Reserved Instance
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/reserved-instances-types.html
upvoted 1 times

ITV2021 1 year, 4 months ago

Selected Answer: A

https://aws.amazon.com/savingsplans/compute-pricing/
upvoted 2 times

Mia2009687 1 year, 5 months ago

Selected Answer: A

EC2 Instance Savings Plan cannot change the family.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 2 times

mattcl 1 year, 5 months ago

Anser D: You can use Standard Reserved Instances when you know that you need a specific instance type.
upvoted 1 times

kruasan 1 year, 7 months ago

Selected Answer: A

Savings Plans offer a flexible pricing model that provides savings on AWS usage. You can save up to 72 percent on your AWS compute workloads. Compute Savings Plans provide lower prices on Amazon EC2 instance usage regardless of instance family, size, OS, tenancy, or AWS Region. This also applies to AWS Fargate and AWS Lambda usage. SageMaker Savings Plans provide you with lower prices for your Amazon SageMaker instance usage, regardless of your instance family, size, component, or AWS Region.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 3 times

kruasan 1 year, 7 months ago

With an EC2 Instance Savings Plan, you can change your instance size within the instance family (for example, from c5.xlarge to c5.2xlarge) or the operating system (for example, from Windows to Linux), or move from Dedicated tenancy to Default and continue to receive the discounted rate provided by your EC2 Instance Savings Plan.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 1 times

kruasan 1 year, 7 months ago

The company needs to be able to change the instance family and sizes in the next 6 months based on application popularity and usage.
Therefore EC2 Instance Savings Plan prerequisites are not fulfilled
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #348 *Topic 1*

A company collects data from a large number of participants who use wearable devices. The company stores the data in an Amazon DynamoDB table and uses applications to analyze the data. The data workload is constant and predictable. The company wants to stay at or below its forecasted budget for DynamoDB.

Which solution will meet these requirements MOST cost-effectively?

A. Use provisioned mode and DynamoDB Standard-Infrequent Access (DynamoDB Standard-IA). Reserve capacity for the forecasted workload.

B. Use provisioned mode. Specify the read capacity units (RCUs) and write capacity units (WCUs). **Most Voted**

C. Use on-demand mode. Set the read capacity units (RCUs) and write capacity units (WCUs) high enough to accommodate changes in the workload.

D. Use on-demand mode. Specify the read capacity units (RCUs) and write capacity units (WCUs) with reserved capacity.

Correct Answer: *B*

Community vote distribution

B (90%) A (10%)

Comments

everfly Highly Voted 1 year, 9 months ago

Selected Answer: B

The data workload is constant and predictable.

upvoted 11 times

hovnival Highly Voted 1 year, 1 month ago

Selected Answer: B

I think it is not possible to set Read Capacity Units(RCU)/Write Capacity Units(WCU) in on-demand mode.

upvoted 6 times

pentium75 Most Recent 11 months ago

Selected Answer: B

C and D are impossible because you don't set or specify RCUs and WCUs in on-demand mode.

A is wrong because there is no indication of "infrequent access", and "the data workload is constant", there is no different between the current and the "forecasted" workload.
upvoted 4 times

wsdasdasdqwdaw 1 year, 1 month ago

predictable/constant => provisioned mode. On-demand mode is more suitable for workloads that are unpredictable and can vary widely from minute to minute.

The use case is not for Standard-IA which is described here: https://aws.amazon.com/dynamodb/standard-ia/

=> Option B
upvoted 3 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: B

I rule out A because of this 'Standard-Infrequent Access ', clearly the company uses applications to analyze the data.
The data workload is constant and predictable making provisioned mode the best option.
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: A

Option B lacks the cost benefits of Standard-IA.

Option C uses more expensive on-demand pricing.

Option D does not actually allow reserving capacity with on-demand mode.

So option A leverages provisioned mode, Standard-IA, and reserved capacity to meet the requirements in a cost-optimal way.
upvoted 1 times

MrAWSAssociate 1 year, 5 months ago

Selected Answer: A

A is correct!
upvoted 1 times

MrAWSAssociate 1 year, 5 months ago

Sorry, A will not work, since Reserved Capacity can only be used with DynamoDB Standard table class. So, B is right for this case.
upvoted 4 times

UNGMAN 1 year, 8 months ago

Selected Answer: B

□□□□..
upvoted 4 times

kayodea25 1 year, 8 months ago

Option C is the most cost-effective solution for this scenario. In on-demand mode, DynamoDB automatically scales up or down based on the current workload, so the company only pays for the capacity it uses. By setting the RCUs and WCUs high enough to accommodate changes in the workload, the company can ensure that it always has the necessary capacity without overprovisioning and incurring unnecessary costs. Since the workload is constant and predictable, using provisioned mode with reserved capacity (Options A and D) may result in paying for unused capacity during periods of low demand. Option B, using provisioned mode without reserved capacity, may result in throttling during periods of high demand if the provisioned capacity is not sufficient to handle the workload.
upvoted 3 times

pentium75 11 months ago

You can't 'set RCUs and WCUs' in on-demand mode.
upvoted 2 times

boxu03 1 year, 8 months ago

you forgot "The data workload is constant and predictable", should be B

you forgot "The data workload is constant and predictable", should be B
upvoted 3 times

Bofi 1 year, 8 months ago

Kayode olode..lol
upvoted 1 times

Steve_4542636 1 year, 9 months ago

"The data workload is constant and predictable."
https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html

"With provisioned capacity you pay for the provision of read and write capacity units for your DynamoDB tables. Whereas with DynamoDB on-demand you pay per request for the data reads and writes that your application performs on your tables."
upvoted 2 times

Charly0710 1 year, 9 months ago

Selected Answer: B

The data workload is constant and predictable, then, isn't on-demand mode.
DynamoDB Standard-IA is not necessary in this context
upvoted 2 times

Lonojack 1 year, 9 months ago

Selected Answer: B

The problem with (A) is: "Standard-Infrequent Access". In the question, they say the company has to analyze the Data.
That's why the Correct answer is (B)
upvoted 4 times

bdp123 1 year, 9 months ago

Selected Answer: A

workload is constant
upvoted 2 times

Lonojack 1 year, 9 months ago

The problem with (A) is: "Standard-Infrequent Access".
In the question, they say the company has to analyze the Data.
Correct answer is (B)
upvoted 3 times

Samuel03 1 year, 9 months ago

Selected Answer: B

As the numbers are already known
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #349 *Topic 1*

A company stores confidential data in an Amazon Aurora PostgreSQL database in the ap-southeast-3 Region. The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key. The company was recently acquired and must securely share a backup of the database with the acquiring company's AWS account in ap-southeast-3.

What should a solutions architect do to meet these requirements?

A. Create a database snapshot. Copy the snapshot to a new unencrypted snapshot. Share the new snapshot with the acquiring company's AWS account.

B. Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account. **Most Voted**

C. Create a database snapshot that uses a different AWS managed KMS key. Add the acquiring company's AWS account to the KMS key alias. Share the snapshot with the acquiring company's AWS account.

D. Create a database snapshot. Download the database snapshot. Upload the database snapshot to an Amazon S3 bucket. Update the S3 bucket policy to allow access from the acquiring company's AWS account.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

Abrar2022 Highly Voted 1 year, 12 months ago

Selected Answer: B

A. - "So let me get this straight, with the current company the data is protected and encrypted. However, for the acquiring company the data is unencrypted? How is that fair?"

C - Wouldn't recommended this option because using a different AWS managed KMS key will not allow the acquiring company's AWS account to access the encrypted data.

D. - Don't risk it for a biscuit and get fired!!!! - by downloading the database snapshot and uploading it to an Amazon S3 bucket. This will increase the risk of data leakage or loss of confidentiality during the transfer process.

B - CORRECT

upvoted 15 times

njufi Most Recent 1 year, 2 months ago

I believe the reason why option C is not the correct answer is that adding the acquiring company's AWS account to the KMS key alias doesn't directly control access to the encrypted data. KMS key aliases are simply alternative names for KMS keys and do not affect access control. Access to encrypted data is goverened by KMS key policies, which define who can use the key for encryption and decryption.

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: B

Create a database snapshot. Add the acquiring company’s AWS account to the KMS key policy. Share the snapshot with the acquiring company’s AWS account.

upvoted 2 times

Vuuu 1 year, 10 months ago

Selected Answer: B

B. Create a database snapshot. Add the acquiring company’s AWS account to the KMS key policy. Share the snapshot with the acquiring company’s AWS account. Most Voted

upvoted 2 times

Abrar2022 1 year, 12 months ago

Create a database snapshot of the encrypted. Add the acquiring company’s AWS account to the KMS key policy. Share the snapshot with the acquiring company’s AWS account.

upvoted 2 times

SkyZeroZx 2 years, 1 month ago

Selected Answer: B

To securely share a backup of the database with the acquiring company's AWS account in the same Region, a solutions architect should create a database snapshot, add the acquiring company's AWS account to the AWS KMS key policy, and share the snapshot with the acquiring company's AWS account.

Option A, creating an unencrypted snapshot, is not recommended as it will compromise the confidentiality of the data. Option C, creating a snapshot that uses a different AWS managed KMS key, does not provide any additional security and will unnecessarily complicate the solution. Option D, downloading the database snapshot and uploading it to an S3 bucket, is not secure as it can expose the data during transit.

Therefore, the correct option is B: Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.

upvoted 2 times

JA2018 6 months, 2 weeks ago

KMS key aliases are simply alternative names for KMS keys and do not affect access control.

Access to encrypted data is governed by KMS key policies, that define who can use the key for encryption and decryption.

upvoted 1 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

Option B is the correct answer.
Option A is not recommended because copying the snapshot to a new unencrypted snapshot will compromise the confidentiality of the data.
Option C is not recommended because using a different AWS managed KMS key will not allow the acquiring company's AWS account to access the encrypted data.
Option D is not recommended because downloading the database snapshot and uploading it to an Amazon S3 bucket will increase the risk of data leakage or loss of confidentiality during the transfer process.

upvoted 3 times

Steve_4542636 2 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html

upvoted 3 times

geekgirl22 2 years, 3 months ago

It is C, you have to create a new key. Read below
You can't share a snapshot that's encrypted with the default AWS KMS key. You must create a custom AWS KMS key instead. To share an encrypted Aurora DB cluster snapshot:

Create a custom AWS KMS key.
Add the target account to the custom AWS KMS key.
Create a copy of the DB cluster snapshot using the custom AWS KMS key. Then, share the newly copied snapshot with the target account.
Copy the shared DB cluster snapshot from the target account
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
upvoted 1 times

leoattf 2 years, 3 months ago

I also thought straight away that it could be C, however, the questions mentions that the database is encrypted with an AWS KMS custom key already. So maybe the letter B could be right, since it already has a custom key, not the default KMS Key. What do you think?
upvoted 3 times

enzomv 2 years, 2 months ago

It is B.
There's no need to create another custom AWS KMS key.
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
Give target account access to the custom AWS KMS key within the source account
1. Log in to the source account, and go to the AWS KMS console in the same Region as the DB cluster snapshot.
2. Select Customer-managed keys from the navigation pane.
3. Select your custom AWS KMS key (ALREADY CREATED)
4. From the Other AWS accounts section, select Add another AWS account, and then enter the AWS account number of your target account.

Then:
Copy and share the DB cluster snapshot
upvoted 3 times

KZM 2 years, 3 months ago

Yes, as per the given information "The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key", it may not be the default AWS KMS key.
upvoted 1 times

KZM 2 years, 3 months ago

Yes, can't share a snapshot that's encrypted with the default AWS KMS key.
But as per the given information "The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key", it may not be the default AWS KMS key.
upvoted 3 times

enzomv 2 years, 2 months ago

I agree with KZM.
It is B.
There's no need to create another custom AWS KMS key.
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
Give target account access to the custom AWS KMS key within the source account
1. Log in to the source account, and go to the AWS KMS console in the same Region as the DB cluster snapshot.
2. Select Customer-managed keys from the navigation pane.
3. Select your custom AWS KMS key (ALREADY CREATED)
4. From the Other AWS accounts section, select Add another AWS account, and then enter the AWS account number of your target account.

Then:
Copy and share the DB cluster snapshot
upvoted 3 times

nyx12345 2 years, 3 months ago

Is it bad that in answer B the acquiring company is using the same KMS key? Should a new KMS key not be used?

upvoted 2 times

geekgirl22 2 years, 3 months ago

Yes, you are right, read my comment above.
upvoted 1 times

bsbs1234 1 year, 8 months ago

I think I would agree with you if option C say using a new "customer managed key" instead of AWS managed key
upvoted 1 times

bdp123 2 years, 3 months ago

Selected Answer: B

https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
upvoted 3 times

jennyka76 2 years, 3 months ago

ANSWER - B
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #350 *Topic 1*

A company uses a 100 GB Amazon RDS for Microsoft SQL Server Single-AZ DB instance in the us-east-1 Region to store customer transactions. The company needs high availability and automatic recovery for the DB instance.

The company must also run reports on the RDS database several times a year. The report process causes transactions to take longer than usual to post to the customers' accounts. The company needs a solution that will improve the performance of the report process.

Which combination of steps will meet these requirements? (Choose two.)

A. Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment. **Most Voted**

B. Take a snapshot of the current DB instance. Restore the snapshot to a new RDS deployment in another Availability Zone.

C. Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica. **Most Voted**

D. Migrate the database to RDS Custom.

E. Use RDS Proxy to limit reporting requests to the maintenance window.

Correct Answer: *AC*

Community vote distribution

AC (100%)

Comments

elearningtakai Highly Voted 2 years, 2 months ago

A and C are the correct choices.
B. It will not help improve the performance of the report process.
D. Migrating to RDS Custom does not address the issue of high availability and automatic recovery.
E. RDS Proxy can help with scalability and high availability but it does not address the issue of performance for the report process. Limiting the reporting requests to the maintenance window will not provide the required availability and recovery for the DB instance.

upvoted 8 times

rockyykrish Most Recent 7 months ago

A and C

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: AC

Create a Multi-AZ deployment, create a read replica of the DB instance in the second Availability Zone, point all requests for reports to the read replica

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: AC

The correct answers are A and C.

A. Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment. This will provide high availability and automatic recovery for the DB instance. If the primary DB instance fails, the standby DB instance will automatically become the primary DB instance. This will ensure that the database is always available.

C. Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica. This will improve the performance of the report process by offloading the read traffic from the primary DB instance to the read replica. The read replica is a fully synchronized copy of the primary DB instance, so the reports will be accurate.

upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: AC

A and C.

upvoted 3 times

WherecanIstart 2 years, 2 months ago

Selected Answer: AC

Options A & C...

upvoted 4 times

KZM 2 years, 3 months ago

Options A+C

upvoted 3 times

bdp123 2 years, 3 months ago

Selected Answer: AC

https://medium.com/awesome-cloud/aws-difference-between-multi-az-and-read-replicas-in-amazon-rds-60fe848ef53a

upvoted 4 times

jennyka76 2 years, 3 months ago

ANSWER - A & C

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #351 *Topic 1*

A company is moving its data management application to AWS. The company wants to transition to an event-driven architecture. The architecture needs to be more distributed and to use serverless concepts while performing the different aspects of the workflow. The company also wants to minimize operational overhead.

Which solution will meet these requirements?

A. Build out the workflow in AWS Glue. Use AWS Glue to invoke AWS Lambda functions to process the workflow steps.

B. Build out the workflow in AWS Step Functions. Deploy the application on Amazon EC2 instances. Use Step Functions to invoke the workflow steps on the EC2 instances.

C. Build out the workflow in Amazon EventBridge. Use EventBridge to invoke AWS Lambda functions on a schedule to process the workflow steps.

D. Build out the workflow in AWS Step Functions. Use Step Functions to create a state machine. Use the state machine to invoke AWS Lambda functions to process the workflow steps. Most Voted

Correct Answer: *D*

Community vote distribution

D (91%) C (9%)

Comments

Lonojack Highly Voted 1 year, 9 months ago

Selected Answer: D

This is why I'm voting D.....QUESTION ASKED FOR IT TO: use serverless concepts while performing the different aspects of the workflow. Is option D utilizing Serverless concepts?

upvoted 13 times

geekgirl22 Highly Voted 1 year, 9 months ago

It is D. Cannot be C because C is "scheduled"

upvoted 8 times

bujuman Most Recent 8 months, 2 weeks ago

Selected Answer: D

While considering this requirement: The architecture needs to be more distributed and to use serverless concepts while

While considering this requirement: The architecture needs to be more distributed and to use serverless concepts while performing the different aspects of the workflow
And checking the following link : https://aws.amazon.com/step-functions/?nc1=h_ls, Answer D is the best for this use case
upvoted 3 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: D

One of the use cases for step functions is to Automate extract, transform, and load (ETL) processes.
https://aws.amazon.com/step-functions/#:~:text=for%20modern%20applications.-,Use%20cases,-Automate%20extract%2C%20transform
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: D

AWS Step functions is serverless Visual workflows for distributed applications
https://aws.amazon.com/step-functions/
upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: D

Step Functions is based on state machines and tasks. A state machine is a workflow. A task is a state in a workflow that represents a single unit of work that another AWS service performs. Each step in a workflow is a state.
Depending on your use case, you can have Step Functions call AWS services, such as Lambda, to perform tasks.
https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html
upvoted 3 times

TariqKipkemei 1 year, 7 months ago

Answer is D.
Step Functions is based on state machines and tasks. A state machine is a workflow. A task is a state in a workflow that represents a single unit of work that another AWS service performs. Each step in a workflow is a state.
Depending on your use case, you can have Step Functions call AWS services, such as Lambda, to perform tasks.
https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html
upvoted 3 times

Karlos99 1 year, 9 months ago

Selected Answer: C

There are two main types of routers used in event-driven architectures: event buses and event topics. At AWS, we offer Amazon EventBridge to build event buses and Amazon Simple Notification Service (SNS) to build event topics.
https://aws.amazon.com/event-driven-architecture/
upvoted 1 times

pentium75 11 months, 1 week ago

How do you 'build out a workflow' in EventBridge?
upvoted 4 times

TungPham 1 year, 9 months ago

Selected Answer: D

Step 3: Create a State Machine
Use the Step Functions console to create a state machine that invokes the Lambda function that you created earlier in Step 1.
https://docs.aws.amazon.com/step-functions/latest/dg/tutorial-creating-lambda-state-machine.html
In Step Functions, a workflow is called a state machine, which is a series of event-driven steps. Each step in a workflow is called a state.
upvoted 3 times

Bilalazure 1 year, 9 months ago

Selected Answer: D

Distrubuted****
upvoted 2 times

Americo32 1 year, 9 months ago

Selected Answer: C

Vou de C, orientada a eventos
upvoted 2 times

MssP 1 year, 8 months ago

It is true that an Event-driven is made with EventBridge but with a Lambda on schedule??? It is a mismatch, isn´t it?
upvoted 2 times

kraken21 1 year, 8 months ago

Tricky question huh!
upvoted 3 times

bdp123 1 year, 9 months ago

Selected Answer: D

AWS Step functions is serverless Visual workflows for distributed applications
https://aws.amazon.com/step-functions/
upvoted 2 times

leoattf 1 year, 9 months ago

Besides, "Visualize and develop resilient workflows for EVENT-DRIVEN architectures."
upvoted 2 times

tellmenowwwww 1 year, 9 months ago

Could it be a C because it's event-driven architecture?
upvoted 3 times

SMAZ 1 year, 9 months ago

Option D..
AWS Step functions are used for distributed applications
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #352 *Topic 1*

A company is designing the network for an online multi-player game. The game uses the UDP networking protocol and will be deployed in eight AWS Regions. The network architecture needs to minimize latency and packet loss to give end users a high-quality gaming experience.

Which solution will meet these requirements?

A. Setup a transit gateway in each Region. Create inter-Region peering attachments between each transit gateway.

B. Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region. Most Voted

C. Set up Amazon CloudFront with UDP turned on. Configure an origin in each Region.

D. Set up a VPC peering mesh between each Region. Turn on UDP for each VPC.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

lucdt4 Highly Voted 2 years ago

Selected Answer: B

AWS Global Accelerator = TCP/UDP minimize latency

upvoted 12 times

OAdekunle Highly Voted 2 years, 1 month ago

General
Q: What is AWS Global Accelerator?

A: AWS Global Accelerator is a networking service that helps you improve the availability and performance of the applications that you offer to your global users. AWS Global Accelerator is easy to set up, configure, and manage. It provides static IP addresses that provide a fixed entry point to your applications and eliminate the complexity of managing specific IP addresses for different AWS Regions and Availability Zones. AWS Global Accelerator always routes user traffic to the optimal endpoint based on performance, reacting instantly to changes in application health, your user's location, and policies that you configure. You can test the performance benefits from your location with a speed comparison tool. Like other AWS services, AWS Global Accelerator is a self-service, pay-per-use offering, requiring no long term commitments or minimum fees.

https://aws.amazon.com/global-accelerator/faqs/

upvoted 5 times

Danilus Most Recent 7 months ago

Selected Answer: B

key-UDP networking protocol
key-minimize latency and packet loss
Transit gateway is to conect multiple vpc
The solution is B because its accept udp protocol and its s global .
The solution is not C because cloudfront just accept http and https protocols

upvoted 2 times

mwwt2022 1 year, 5 months ago

online game -> Global Accelerator
cloudfront is for static/dynamic content caching

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region.

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: B

Connect to up to 10 regions within the AWS global network using the AWS Global Accelerator.

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

UDP = Global Accelerator

upvoted 2 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

Global Accelerator supports the User Datagram Protocol (UDP) and Transmission Control Protocol (TCP), making it an excellent choice for an online multi-player game using UDP networking protocol. By setting up Global Accelerator with UDP listeners and endpoint groups in each Region, the network architecture can minimize latency and packet loss, giving end users a high-quality gaming experience.

upvoted 5 times

Bofi 2 years, 3 months ago

Selected Answer: B

AWS Global Accelerator is a service that improves the availability and performance of applications with local or global users. Global Accelerator improves performance for a wide range of applications over TCP or UDP by proxying packets at the edge to applications running in one or more AWS Regions. Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover. Both services integrate with AWS Shield for DDoS protection.

upvoted 2 times

K0nAn 2 years, 3 months ago

Selected Answer: B

Global Accelerator for UDP and TCP traffic

upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: B

Global Accelerator

upvoted 2 times

Neha999 2 years, 3 months ago

B

B
Global Accelerator for UDP traffic
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #353 *Topic 1*

A company hosts a three-tier web application on Amazon EC2 instances in a single Availability Zone. The web application uses a self-managed MySQL database that is hosted on an EC2 instance to store data in an Amazon Elastic Block Store (Amazon EBS) volume. The MySQL database currently uses a 1 TB Provisioned IOPS SSD (io2) EBS volume. The company expects traffic of 1,000 IOPS for both reads and writes at peak traffic.

The company wants to minimize any disruptions, stabilize performance, and reduce costs while retaining the capacity for double the IOPS. The company wants to move the database tier to a fully managed solution that is highly available and fault tolerant.

Which solution will meet these requirements MOST cost-effectively?

A. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with an io2 Block Express EBS volume.

B. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume. **Most Voted**

C. Use Amazon S3 Intelligent-Tiering access tiers.

D. Use two large EC2 instances to host the database in active-passive mode.

Correct Answer: *B*

Community vote distribution

B (92%) A (8%)

Comments

AlmeroSenior Highly Voted 2 years, 3 months ago

Selected Answer: B

RDS does not support IO2 or IO2express . GP2 can do the required IOPS

RDS supported Storage >
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
GP2 max IOPS >
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/general-purpose.html#gp2-performance

upvoted 16 times

sophieb Highly Voted 1 year, 2 months ago

Selected Answer: B

RDS now supports io2 but it might still be an overkill given Gp2 is enough and we are looking for the most cost effective solution.

upvoted 8 times

EllenLiu Most Recent 5 months, 3 weeks ago

Selected Answer: B

gp2: 1,000 GiB => 3000 Baseline performance (IOPS) match the requirement which is 2000 only
Baseline IOPS performance : 100 ~16,000 at a rate of 3 IOPS per GiB of volume size.

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

RDS does not support IO2 or IO2express . GP2 can do the required IOPS

upvoted 4 times

Gooniegoogoo 1 year, 11 months ago

The Options is A only because it is sufficient.. Provisioned IOPS are available but overkill.. just want to make sure we understand why its A for the right reason

upvoted 1 times

dkw2342 1 year, 3 months ago

Provisioned IOPS are available, but not io2, just io1.

upvoted 1 times

Abrar2022 1 year, 12 months ago

Simplified by Almero - thanks.

RDS does not support IO2 or IO2express . GP2 can do the required IOPS

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: B

I tried on the portal and only gp3 and i01 are supported.
This is 11 May 2023.

upvoted 4 times

ruqui 2 years ago

it doesn't matter whether or no io* is supported, using io2 is overkill, you only need 1K IOPS, B is the correct answer

upvoted 2 times

SimiTik 2 years, 1 month ago

A
Amazon RDS supports the use of Amazon EBS Provisioned IOPS (io2) volumes. When creating a new DB instance or modifying an existing one, you can select the io2 volume type and specify the amount of IOPS and storage capacity required. RDS also supports the newer io2 Block Express volumes, which can deliver even higher performance for mission-critical database workloads.

upvoted 2 times

TariqKipkemei 2 years ago

Impossible. I just tried on the portal and only io1 and gp3 are supported.

upvoted 2 times

klayytech 2 years, 2 months ago

Selected Answer: B

he most cost-effective solution that meets the requirements is to use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume. This solution will provide high availability and fault tolerance while minimizing disruptions and stabilizing performance. The gp2 EBS volume can handle up to 16,000 IOPS. You can also scale up to 64 TiB of storage.

Amazon RDS for MySQL provides automated backups, software patching, and automatic host replacement. It also provides Multi-AZ deployments that automatically replicate data to a standby instance in another Availability Zone. This ensures that data is always available even in the event of a failure.

upvoted 2 times

test_devops_aws 2 years, 2 months ago

Selected Answer: B

RDS does not support io2 !!!

upvoted 1 times

Maximus007 2 years, 2 months ago

B:gp3 would be the better option, but considering we have only gp2 option and such storage volume - gp2 will be the right choice

upvoted 4 times

Nel8 2 years, 2 months ago

Selected Answer: B

I thought the answer here is A. But when I found the link from Amazon website; as per AWS:

Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance and cost to the needs of your database workload. You can create MySQL, MariaDB, Oracle, and PostgreSQL RDS DB instances with up to 64 tebibytes (TiB) of storage. You can create SQL Server RDS DB instances with up to 16 TiB of storage. For this amount of storage, use the Provisioned IOPS SSD and General Purpose SSD storage types.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

upvoted 1 times

JA2018 6 months, 2 weeks ago

[updated - 21 Nove 2024]

Amazon RDS provides three storage types: Provisioned IOPS SSD (also known as io1 and io2 Block Express), General Purpose SSD (also known as gp2 and gp3), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance and cost to the needs of your database workload. You can create Db2, MySQL, MariaDB, Oracle, SQL Server, and PostgreSQL RDS DB instances with up to 64 tebibytes (TiB) of storage. RDS for Db2 doesn't support the gp2 and magnetic storage types.

upvoted 1 times

Steve_4542636 2 years, 3 months ago

Selected Answer: B

for DB instances between 1 TiB and 4 TiB, storage is striped across four Amazon EBS volumes providing burst performance of up to 12,000 IOPS.

from "https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html"

upvoted 2 times

TungPham 2 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1), and magnetic (also known as standard)
B - MOST cost-effectively

upvoted 4 times

KZM 2 years, 3 months ago

The baseline IOPS performance of gp2 volumes is 3 IOPS per GB, which means that a 1 TB gp2 volume will have a baseline performance of 3,000 IOPS. However, the volume can also burst up to 16,000 IOPS for short periods, but this burst performance is limited and may not be sustained for long durations.

So, I am more prefer option A.
upvoted 1 times

KZM 2 years, 3 months ago

If a 1 TB gp3 EBS volume is used, the maximum available IOPS according to calculations is 3000. This means that the storage can support a requirement of 1000 IOPS, and even 2000 IOPS if the requirement is doubled.
I am confusing between choosing A or B.
upvoted 2 times

mark16dc 2 years, 3 months ago

Selected Answer: A

Option A is the correct answer. A Multi-AZ deployment provides high availability and fault tolerance by automatically replicating data to a standby instance in a different Availability Zone. This allows for seamless failover in the event of a primary instance failure. Using an io2 Block Express EBS volume provides the needed IOPS performance and capacity for the database. It is also designed for low latency and high durability, which makes it a good choice for a database tier.
upvoted 1 times

CapJackSparrow 2 years, 2 months ago

How will you select io2 when RDS only offers io1....magic?
upvoted 1 times

bdp123 2 years, 3 months ago

Selected Answer: B

Correction - hit wrong answer button - meant 'B'
Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1)
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #354 *Topic 1*

A company hosts a serverless application on AWS. The application uses Amazon API Gateway, AWS Lambda, and an Amazon RDS for PostgreSQL database. The company notices an increase in application errors that result from database connection timeouts during times of peak traffic or unpredictable traffic. The company needs a solution that reduces the application failures with the least amount of change to the code.

What should a solutions architect do to meet these requirements?

A. Reduce the Lambda concurrency rate.

B. Enable RDS Proxy on the RDS DB instance. Most Voted

C. Resize the RDS DB instance class to accept more connections.

D. Migrate the database to Amazon DynamoDB with on-demand scaling.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

TariqKipkemei Highly Voted 2 years ago

Selected Answer: B

Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability. With RDS Proxy, failover times for Aurora and RDS databases are reduced by up to 66%.

https://aws.amazon.com/rds/proxy/

upvoted 11 times

Murtadhaceit Highly Voted 1 year, 6 months ago

Selected Answer: B

A. Reduce the Lambda concurrency rate? Has nothing to do with decreasing connections timeout.
B. Enable RDS Proxy on the RDS DB instance. Correct answer
C. Resize the RDS DB instance class to accept more connections? More connections means worse performance. Therefore, not correct

correct.
D. Migrate the database to Amazon DynamoDB with on-demand scaling? DynamoDB is a noSQL database. Not correct.
upvoted 6 times

JA2018 Most Recent 6 months, 2 weeks ago

Selected Answer: B

Keys found in STEM:

1. The company notices an increase in application errors that result from database connection timeouts during times of peak traffic or unpredictable traffic.

2. The company needs a solution that reduces the application failures with the least amount of change to the code.
upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

RDS Proxy is a fully managed, highly available, and scalable proxy for Amazon Relational Database Service (RDS) that makes it easy to connect to your RDS instances from applications running on AWS Lambda. RDS Proxy offloads the management of connections to the database, which can help to improve performance and reliability.
upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

To reduce application failures resulting from database connection timeouts, the best solution is to enable RDS Proxy on the RDS DB instance
upvoted 2 times

WherecanIstart 2 years, 2 months ago

Selected Answer: B

RDS Proxy
upvoted 4 times

nder 2 years, 3 months ago

Selected Answer: B

RDS Proxy will pool connections, no code changes need to be made
upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: B

RDS proxy
upvoted 2 times

Neha999 2 years, 3 months ago

B RDS Proxy
https://aws.amazon.com/rds/proxy/
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #355 *Topic 1*

A company is migrating an old application to AWS. The application runs a batch job every hour and is CPU intensive. The batch job takes 15 minutes on average with an on-premises server. The server has 64 virtual CPU (vCPU) and 512 GiB of memory.

Which solution will run the batch job within 15 minutes with the LEAST operational overhead?

A. Use AWS Lambda with functional scaling.

B. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate.

C. Use Amazon Lightsail with AWS Auto Scaling.

D. Use AWS Batch on Amazon EC2. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (98%) A

Comments

NolaHOla Highly Voted 2 years, 3 months ago

The amount of CPU and memory resources required by the batch job exceeds the capabilities of AWS Lambda and Amazon Lightsail with AWS Auto Scaling, which offer limited compute resources. AWS Fargate offers containerized application orchestration and scalable infrastructure, but may require additional operational overhead to configure and manage the environment. AWS Batch is a fully managed service that automatically provisions the required infrastructure for batch jobs, with options to use different instance types and launch modes.

Therefore, the solution that will run the batch job within 15 minutes with the LEAST operational overhead is D. Use AWS Batch on Amazon EC2. AWS Batch can handle all the operational aspects of job scheduling, instance management, and scaling while using Amazon EC2 injavascript:void(0)stances with the right amount of CPU and memory resources to meet the job's requirements.

upvoted 22 times

everfly Highly Voted 2 years, 3 months ago

Selected Answer: D

AWS Batch is a fully-managed service that can launch and manage the compute resources needed to execute batch jobs. It can scale the compute environment based on the size and timing of the batch jobs.

upvoted 12 times

FlyingHawk Most Recent 4 months, 2 weeks ago

Selected Answer: D

Lambda has a hard limit of 10 GB of memory and 6 vCPUs per function execution, Amazon Light sail has the limit of 8vCPUs and 32GB memory, so A and C out, B and D can meet vCPU and memory requirements, but B is more expensive than D, so correct answer is D.

upvoted 1 times

Danilus 7 months ago

Selected Answer: D

key-The batch job takes 15 minutes on average
key-LEAST operational overhead
key-the batch job
is not A because lambda execution takes 15 max and lambda dont support 512 GIB of memory
the answer is D batch it is designed to run large-scale batch jobs an automatically manages the scaling of resources also batch allow jobs to be distributed accross multiple instances because supports parallel execution

upvoted 3 times

Ramdi1 1 year, 8 months ago

Selected Answer: D

The question needs to be phrased differently. I assume at first it was Lambda, because it says 15 minutes in the question which can be done. Yes it also does say CPU intensive however they go on with a full stop and then give you the server specs. It does not say it uses that much of the specs so they need to really rephrase the questions.

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

The main reasons are:

AWS Batch can easily schedule and run batch jobs on EC2 instances. It can scale up to the required vCPUs and memory to match the on-premises server.
Using EC2 provides full control over the instance type to meet the resource needs.
No servers or clusters to manage like with ECS/Fargate or Lightsail. AWS Batch handles this automatically.
More cost effective and operationally simple compared to Lambda which is not ideal for long running batch jobs.

upvoted 7 times

BrijMohan08 1 year, 9 months ago

Selected Answer: A

On-Prem was avg 15 min, but target state architecture is expected to finish within 15 min

upvoted 1 times

pentium75 1 year, 5 months ago

How? The on-prem server has 64 CPUs and 512 GB RAM, Lambda offers much less. And even on-prem it takes 15 minutes ON AVERAGE, sometimes more.

upvoted 5 times

jayce5 1 year, 10 months ago

Selected Answer: D

Not Lambda, "average 15 minutes" means there are jobs with running more and less than 15 minutes. Lambda max is 15 minutes.

upvoted 3 times

Gooniegoogoo 1 year, 11 months ago

This is for certain a tough one. I do see that they have thrown a curve ball in making it Lambda Functional scaling, however what we dont know is if this application has many request or one large one.. looks like Lambda can scale and use the same lambda env.. seems intensive tho so will go with D

upvoted 5 times

TariqKipkemei 2 years ago

Selected Answer: D

AWS Batch

upvoted 3 times

JLII 2 years, 3 months ago

Selected Answer: D

Not A because: "AWS Lambda now supports up to 10 GB of memory and 6 vCPU cores for Lambda Functions." https://aws.amazon.com/about-aws/whats-new/2020/12/aws-lambda-supports-10gb-memory-6-vcpu-cores-lambda-functions/ vs. "The server has 64 virtual CPU (vCPU) and 512 GiB of memory" in the question.

upvoted 7 times

geekgirl22 2 years, 3 months ago

A is the answer. Lambda is known that has a limit of 15 minutes. So for as long as it says "within 15 minutes" that should be a clear indication it is Lambda

upvoted 2 times

nder 2 years, 3 months ago

Wrong, the job takes "On average 15 minutes" and requires more cpu and ram than lambda can deal with. AWS Batch is correct in this scenario

upvoted 4 times

geekgirl22 2 years, 3 months ago

read the rest of the question which gives the answer:
"Which solution will run the batch job within 15 minutes with the LEAST operational overhead?"
Keyword "Within 15 minutes"

upvoted 3 times

Lonojack 2 years, 3 months ago

What happens if it EXCEEDS the 15 min AVERAGE?
Average = possibly can be more than 15min.
The safer bet would be option D: AWS Batch on EC2

upvoted 8 times

Terion 1 year, 8 months ago

I think what he means is that it takes on average 15 min on prem only

upvoted 2 times

awsgeek75 1 year, 4 months ago

How are you going to get 64 vCPUS to a Lambda function?

upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: D

AWS batch on EC2

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #356 *Topic 1*

A company stores its data objects in Amazon S3 Standard storage. A solutions architect has found that 75% of the data is rarely accessed after 30 days. The company needs all the data to remain immediately accessible with the same high availability and resiliency, but the company wants to minimize storage costs.

Which storage solution will meet these requirements?

A. Move the data objects to S3 Glacier Deep Archive after 30 days.

B. Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days. Most Voted

C. Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.

D. Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

Lonojack Highly Voted 2 years, 3 months ago

Selected Answer: B

Needs immediate accessibility after 30days, IF they need to be accessed.

upvoted 6 times

Piccalo Highly Voted 2 years, 2 months ago

Highly available so One Zone IA is out the question
Glacier Deep archive isn't immediately accessible 12-48 hours
B is the answer.

upvoted 5 times

Danilus Most Recent 7 months ago

Selected Answer: B

Key - high availability and resiliency
Key-the data to remain immediately accessible

Its not C or D because its One zone

its no B even if glacier is cheaper the problem is that data must be immediately accesible so the answer is B

upvoted 2 times

Aru_1994 10 months, 3 weeks ago

Option B

upvoted 2 times

Apexakil1996 1 year, 5 months ago

One -zone -infrequent access cannot be the answer because it requires high availability so standard infrequent access should be the answer

upvoted 5 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: B

high availability, resiliency = multi AZ
75% of the data is rarely accessed but remain immediately accessible = Standard-Infrequent Access

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

The correct answer is B.

S3 Standard-IA is a storage class that is designed for infrequently accessed data. It offers lower storage costs than S3 Standard, but it has a retrieval latency of 1-5 minutes.

upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

S3 Glacier Deep Archive is intended for data that is rarely accessed and can tolerate retrieval times measured in hours. Moving data to S3 One Zone-IA immediately would not meet the requirement of immediate accessibility with the same high availability and resiliency.

upvoted 2 times

KS2020 2 years, 2 months ago

The answer should be C.
S3 One Zone-IA is for data that is accessed less frequently but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZs), S3 One Zone-IA stores data in a single AZ and costs 20% less than S3 Standard-IA.

https://aws.amazon.com/s3/storage-classes/#:~:text=S3%20One%20Zone%2DIA%20is,less%20than%20S3%20Standard%2DIA.

upvoted 1 times

shanwford 2 years, 2 months ago

The Question emphasises to kepp same high availability class - S3 One Zone-IA doesnt support multiple Availability Zone data resilience model like S3 Standard-Infrequent Access.

upvoted 3 times

bdp123 2 years, 3 months ago

Selected Answer: B

S3 Standard-Infrequent Access after 30 days

upvoted 3 times

NolaHOla 2 years, 3 months ago

B
Option B - Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days - will meet the requirements of keeping the data immediately accessible with high availability and resiliency, while minimizing storage costs. S3 Standard-IA is designed for infrequently accessed data, and it provides a lower storage cost than S3 Standard, while still offering the same low latency, high throughput, and high durability as S3 Standard.

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #357 *Topic 1*

A gaming company is moving its public scoreboard from a data center to the AWS Cloud. The company uses Amazon EC2 Windows Server instances behind an Application Load Balancer to host its dynamic application. The company needs a highly available storage solution for the application. The application consists of static files and dynamic server-side code.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

A. Store the static files on Amazon S3. Use Amazon CloudFront to cache objects at the edge. **Most Voted**

B. Store the static files on Amazon S3. Use Amazon ElastiCache to cache objects at the edge.

C. Store the server-side code on Amazon Elastic File System (Amazon EFS). Mount the EFS volume on each EC2 instance to share the files.

D. Store the server-side code on Amazon FSx for Windows File Server. Mount the FSx for Windows File Server volume on each EC2 instance to share the files. **Most Voted**

E. Store the server-side code on a General Purpose SSD (gp2) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on each EC2 instance to share the files.

Correct Answer: *AD*

Community vote distribution

AD (100%)

Comments

Steve_4542636 Highly Voted 2 years, 3 months ago

Selected Answer: AD

A because Elasticache, despite being ideal for leaderboards per Amazon, doesn't cache at edge locations. D because FSx has higher performance for low latency needs.

https://www.techtarget.com/searchaws/tip/Amazon-FSx-vs-EFS-Compare-the-AWS-file-services

"FSx is built for high performance and submillisecond latency using solid-state drive storage volumes. This design enables users to select storage capacity and latency independently. Thus, even a subterabyte file system can have 256 Mbps or higher throughput and support volumes up to 64 TB."

upvoted 7 times

Nel8 2 years, 2 months ago

Just to add, ElastiCache is use in front of AWS database.
upvoted 2 times

baba365 1 year, 8 months ago

Why not EFS?
upvoted 1 times

Guru4Cloud Highly Voted 1 year, 9 months ago

Selected Answer: AD

The reasons are:

Storing static files in S3 with CloudFront provides durability, high availability, and low latency by caching at edge locations.
FSx for Windows File Server provides a fully managed Windows native file system that can be accessed from the Windows EC2 instances to share server-side code. It is designed for high availability and scales up to 10s of GBPS throughput.
EFS and EBS volumes can be attached to a single AZ. FSx and S3 are replicated across AZs for high availability.
upvoted 6 times

Danilus Most Recent 7 months ago

Selected Answer: AD

Key: Static files and dynamic server-side code.
Key: Windows Server.

The answer is A because S3 is for static files, and CloudFront is a CDN.

The second answer is D because FSx works for Windows File Server.

It's not C because EFS is just for Linux.
It's not E because EBS doesn't share files between EC2 instances.
upvoted 2 times

rodrigoleoncio 1 year ago

Selected Answer: AD

A because Elasticache doesn't cache at edge locations. D because FSx has higher performance for low latency needs.
upvoted 2 times

awsgeek75 1 year, 4 months ago

The question and options are badly worded. How does (D) storing server side code on a file server makes it executable?
upvoted 2 times

4fad2f8 1 year, 4 months ago

you can't mount efs on windows
upvoted 3 times

WherecanIstart 2 years, 2 months ago

Selected Answer: AD

A & D for sure
upvoted 5 times

KZM 2 years, 3 months ago

It is obvious that A and D.
upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: AD

both A and D seem correct
upvoted 2 times

NolaHOla 2 years, 3 months ago

A and D seems correct

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #358 *Topic 1*

A social media company runs its application on Amazon EC2 instances behind an Application Load Balancer (ALB). The ALB is the origin for an Amazon CloudFront distribution. The application has more than a billion images stored in an Amazon S3 bucket and processes thousands of images each second. The company wants to resize the images dynamically and serve appropriate formats to clients.

Which solution will meet these requirements with the LEAST operational overhead?

A. Install an external image management library on an EC2 instance. Use the image management library to process the images.

B. Create a CloudFront origin request policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

C. Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors that serve the images. **Most Voted**

D. Create a CloudFront response headers policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

Correct Answer: *C*

Community vote distribution

C (88%) B (12%)

Comments

NolaHOla Highly Voted 1 year, 9 months ago

Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors that serve the images.

Using a Lambda@Edge function with an external image management library is the best solution to resize the images dynamically and serve appropriate formats to clients. Lambda@Edge is a serverless computing service that allows running custom code in response to CloudFront events, such as viewer requests and origin requests. By using a Lambda@Edge function, it's possible to process images on the fly and modify the CloudFront response before it's sent back to the client. Additionally, Lambda@Edge has built-in support for external libraries that can be used to process images. This approach will reduce operational overhead and scale automatically with traffic.

upvoted 21 times

TariqKipkemei Highly Voted 1 year, 1 month ago

Selected Answer: C

The moment there is a need to implement some logic at the CDN think Lambda@Edge.
upvoted 10 times

Guru4Cloud Most Recent 1 year, 3 months ago

Selected Answer: C

The correct answer is C.

A Lambda@Edge function is a serverless function that runs at the edge of the CloudFront network. This means that the function is executed close to the user, which can improve performance.
An external image management library can be used to resize images and to serve the appropriate format.
Associating the Lambda@Edge function with the CloudFront behaviors that serve the images ensures that the function is executed for all requests that are served by those behaviors.
upvoted 4 times

BrijMohan08 1 year, 3 months ago

Selected Answer: B

If the user asks for the most optimized image format (JPEG,WebP, or AVIF) using the directive format=auto, CloudFront Function will select the best format based on the Accept header present in the request.

Latest documentation: https://aws.amazon.com/blogs/networking-and-content-delivery/image-optimization-using-amazon-cloudfront-and-aws-lambda/
upvoted 3 times

pentium75 11 months, 1 week ago

But a policy alone cannot resize images.
upvoted 1 times

bdp123 1 year, 9 months ago

Selected Answer: C

https://aws.amazon.com/cn/blogs/networking-and-content-delivery/resizing-images-with-amazon-cloudfront-lambdaedge-aws-cdn-blog/
upvoted 5 times

everfly 1 year, 9 months ago

Selected Answer: C

https://aws.amazon.com/cn/blogs/networking-and-content-delivery/resizing-images-with-amazon-cloudfront-lambdaedge-aws-cdn-blog/
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #359 *Topic 1*

A hospital needs to store patient records in an Amazon S3 bucket. The hospital's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest. The compliance team must administer the encryption key for data at rest.

Which solution will meet these requirements?

A. Create a public SSL/TLS certificate in AWS Certificate Manager (ACM). Associate the certificate with Amazon S3. Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

B. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with S3 managed encryption keys (SSE-S3). Assign the compliance team to manage the SSE-S3 keys.

C. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys. **Most Voted**

D. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Use Amazon Macie to protect the sensitive data that is stored in Amazon S3. Assign the compliance team to manage Macie.

Correct Answer: *C*

Community vote distribution

C (89%) Other (11%)

Comments

NolaHOla Highly Voted 1 year, 9 months ago

Option C is correct because it allows the compliance team to manage the KMS keys used for server-side encryption, thereby providing the necessary control over the encryption keys. Additionally, the use of the "aws:SecureTransport" condition on the bucket policy ensures that all connections to the S3 bucket are encrypted in transit.
option B might be misleading but using SSE-S3, the encryption keys are managed by AWS and not by the compliance team
upvoted 29 times

Lonojack 1 year, 9 months ago

Perfect explanation. I Agree
upvoted 5 times

pentium75 Highly Voted 11 months, 1 week ago

Selected Answer: C

Not A, Certificate Manager has nothing to do with S3
Not B, SSE-S3 does not allow compliance team to manage the key
Not D, Macie is for identifying sensitive data, not protecting it
upvoted 13 times

Guru4Cloud Most Recent 1 year, 3 months ago

Selected Answer: C

Macie does not encrypt the data like the question is asking
https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html

Also, SSE-S3 encryption is fully managed by AWS so the Compliance Team can't administer this.
upvoted 3 times

Yadav_Sanjay 1 year, 6 months ago

Selected Answer: C

D - Can't be because - Amazon Macie is a data security service that uses machine learning (ML) and pattern matching to discover and help protect your sensitive data.
Macie discovers sensitive information, can help in protection but can't protect
upvoted 3 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: C

B can work if they do not want control over encryption keys.
upvoted 2 times

Russs99 1 year, 8 months ago

Selected Answer: A

Option A proposes creating a public SSL/TLS certificate in AWS Certificate Manager and associating it with Amazon S3. This step ensures that data is encrypted in transit. Then, the default encryption for each S3 bucket will be configured to use server-side encryption with AWS KMS keys (SSE-KMS), which will provide encryption at rest for the data stored in S3. In this solution, the compliance team will manage the KMS keys, ensuring that they control the encryption keys for data at rest.
upvoted 1 times

pentium75 11 months, 1 week ago

ACM is for website certificates, has nothing to do with S3.
upvoted 3 times

Shrestwt 1 year, 7 months ago

ACM cannot be integrated with Amazon S3 bucket directly.
upvoted 3 times

Bofi 1 year, 8 months ago

Selected Answer: C

Option C seems to be the correct answer, option A is also close but ACM cannot be integrated with Amazon S3 bucket directly, hence, u can not attached TLS to S3. You can only attached TLS certificate to ALB, API Gateway and CloudFront and maybe Global Accelerator but definitely NOT EC2 instance and S3 bucket
upvoted 2 times

CapJackSparrow 1 year, 8 months ago

Selected Answer: C

D makes no sense.
upvoted 3 times

Dody 1 year, 9 months ago

Selected Answer: C

Correct Answer is "C"
“D” is not correct because Amazon Macie securely stores your data at rest using AWS encryption solutions. Macie encrypts data, such as findings, using an AWS managed key from AWS Key Management Service (AWS KMS). However, in the question there is a requirement that the compliance team must administer the encryption key for data at rest.
https://docs.aws.amazon.com/macie/latest/user/data-protection.html

upvoted 3 times

cegama543 1 year, 9 months ago

Selected Answer: C

Option C will meet the requirements.

Explanation:

The compliance team needs to administer the encryption key for data at rest in order to ensure that protected health information (PHI) is encrypted in transit and at rest. Therefore, we need to use server-side encryption with AWS KMS keys (SSE-KMS). The default encryption for each S3 bucket can be configured to use SSE-KMS to ensure that all new objects in the bucket are encrypted with KMS keys.

Additionally, we can configure the S3 bucket policies to allow only encrypted connections over HTTPS (TLS) using the aws:SecureTransport condition. This ensures that the data is encrypted in transit.

upvoted 2 times

Karlos99 1 year, 9 months ago

Selected Answer: C

We must provide encrypted in transit and at rest. Macie is needed to discover and recognize any PII or Protected Health Information. We already know that the hospital is working with the sensitive data) so protect them witn KMS and SSL. Answer D is unnecessary

upvoted 2 times

Steve_4542636 1 year, 9 months ago

Selected Answer: C

Macie does not encrypt the data like the question is asking
https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html

Also, SSE-S3 encryption is fully managed by AWS so the Compliance Team can't administer this.

upvoted 3 times

Abhineet9148232 1 year, 9 months ago

Selected Answer: C

C [Correct]: Ensures Https only traffic (encrypted transit), Enables compliance team to govern encryption key.
D [Incorrect]: Misleading; PHI is required to be encrypted not discovered. Maice is a discovery service. (https://aws.amazon.com/macie/)

upvoted 5 times

Nel8 1 year, 9 months ago

Selected Answer: D

Correct answer should be D. "Use Amazon Macie to protect the sensitive data..."
As requirement says "The hospitals's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest."

Macie protects personal record such as PHI. Macie provides you with an inventory of your S3 buckets, and automatically evaluates and monitors the buckets for security and access control. If Macie detects a potential issue with the security or privacy of your data, such as a bucket that becomes publicly accessible, Macie generates a finding for you to review and remediate as necessary.

upvoted 4 times

Drayen25 1 year, 9 months ago

Option C should be

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #360 *Topic 1*

A company uses Amazon API Gateway to run a private gateway with two REST APIs in the same VPC. The BuyStock RESTful web service calls the CheckFunds RESTful web service to ensure that enough funds are available before a stock can be purchased. The company has noticed in the VPC flow logs that the BuyStock RESTful web service calls the CheckFunds RESTful web service over the internet instead of through the VPC. A solutions architect must implement a solution so that the APIs communicate through the VPC.

Which solution will meet these requirements with the FEWEST changes to the code?

A. Add an X-API-Key header in the HTTP header for authorization.

B. Use an interface endpoint. **Most Voted**

C. Use a gateway endpoint.

D. Add an Amazon Simple Queue Service (Amazon SQS) queue between the two REST APIs.

Correct Answer: *B*

Community vote distribution

B (94%) C (6%)

Comments

everfly Highly Voted 1 year, 9 months ago

Selected Answer: B

an interface endpoint is a horizontally scaled, redundant VPC endpoint that provides private connectivity to a service. It is an elastic network interface with a private IP address that serves as an entry point for traffic destined to the AWS service. Interface endpoints are used to connect VPCs with AWS services

upvoted 23 times

lucdt4 Highly Voted 1 year, 6 months ago

Selected Answer: B

C. Use a gateway endpoint is wrong because gateway endpoints only support for S3 and dynamoDB, so B is correct

upvoted 12 times

zdi561 Most Recent 4 months ago

Selected Answer: B

AWS PrivateLink connects resources within one VPC (the consumer) to services within another VPC (the provider) using private IP addresses. But since it uses API gateway, you have to use VPC endpoint to keep private communication though there are other options to avoid using VPC endpoint

upvoted 1 times

Rcosmos 4 months, 2 weeks ago

Selected Answer: B

B. Usar um endpoint de interface:

Embora os endpoints de interface também possam facilitar a comunicação privada, eles são projetados para integrar serviços AWS ou serviços baseados em VPC endpoints. Para o caso de APIs REST privadas do API Gateway, o endpoint de gateway é mais apropriado.

upvoted 1 times

Rcosmos 4 months, 2 weeks ago

Selected Answer: B

Por que o endpoint de gateway é a solução ideal?
Comunicação por meio da VPC:Um endpoint de gateway no Amazon API Gateway permite que os serviços na mesma VPC se comuniquem diretamente, sem sair da rede interna ou passar pela Internet pública.
Menor impacto no código:O uso de um endpoint de gateway no API Gateway requer apenas a reconfiguração do endpoint na VPC, sem mudanças significativas no código das APIs.
Segurança:Com um endpoint de gateway, o tráfego é mantido dentro da rede privada da VPC, reduzindo os riscos de segurança associados ao tráfego pela Internet.
Custo-benefício:Um endpoint de gateway é econômico, pois utiliza a infraestrutura existente da VPC e evita custos adicionais relacionados ao tráfego pela Internet.

upvoted 1 times

meowruki 1 year ago

Selected Answer: B

B. Use an interface endpoint.

Here's the reasoning:

Interface Endpoint (Option B): An interface endpoint (also known as VPC endpoint) allows communication between resources in your VPC and services without traversing the public internet. In this case, you can create an interface endpoint for API Gateway in your VPC. This enables the communication between the BuyStock and CheckFunds RESTful web services within the VPC, and it doesn't require significant changes to the code.

X-API-Key header (Option A): Adding an X-API-Key header for authorization doesn't address the issue of ensuring that the APIs communicate through the VPC. It's more related to authentication and authorization mechanisms.

upvoted 5 times

liux99 1 year, 1 month ago

The question here is that the BuyStock RESTful web service calls the CheckFunds RESTful web service through API gateway (internet), not directly. How does API gateway connect the services BuyStock and CheckFunds? It connects the Interface Endpoint of the services through Privatelink. The interface endpoints provide direct connection between services within the same private subnet. Answer B is correct.

upvoted 3 times

youdelin 1 year, 1 month ago

how is it even possible, I mean if it's private and both are in the same VPC then we shouldn't even have such an issue right?

upvoted 4 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: B

B. Use an interface endpoint.

upvoted 2 times

envest 1 year, 6 months ago

Answer B (from abylead)
With API GW, you can create multiple prv REST APIs, only accessible with an interface VPC endpt. To allow/ deny simple or cross acc access to your API from selected VPCs & its endpts, you use resource plcys. In addition, you can also use DX for a connection between onprem network to VPC or your prv API

connection between onprem network to VPC or your priv API.
API GW to VPC: https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-private-apis.html

Less correct & incorrect (infeasible & inadequate) answers:
A)X-API-Key in HTTP header for authorization needs auto-process fcts & changes: inadequate.
C)VPC GW endpts for S3 or DynamDB aren't for RESTful svcs: infeasible.
D)SQS que between 2 REST APIs needs endpts & some changes: inadequate.
upvoted 2 times

aqmdla2002 1 year, 6 months ago

Selected Answer: C

I select C because it's the solution with the " FEWEST changes to the code"
upvoted 1 times

pentium75 11 months, 1 week ago

Gateway Endpoint can provide access to S3 or DynamoDB, not to API Gateway
upvoted 2 times

awsgeek75 10 months, 3 weeks ago

Fewest changes to the code doesn't mean break the code by doing something irrelevant. Gateway endpoint is for S3 and DynamoDB
upvoted 2 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: B

An interface endpoint is powered by PrivateLink, and uses an elastic network interface (ENI) as an entry point for traffic destined to the service
upvoted 3 times

kprakashbehera 1 year, 8 months ago

Selected Answer: B

BBBBBB
upvoted 2 times

siyam008 1 year, 9 months ago

Selected Answer: C

https://www.linkedin.com/pulse/aws-interface-endpoint-vs-gateway-alex-chang
upvoted 1 times

siyam008 1 year, 9 months ago

Correct answer is B. Incorrectly selected C
upvoted 3 times

DASBOL 1 year, 9 months ago

Selected Answer: B

https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-private-apis.html
upvoted 5 times

Sherif_Abbas 1 year, 9 months ago

Selected Answer: C

The only time where an Interface Endpoint may be preferable (for S3 or DynamoDB) over a Gateway Endpoint is if you require access from on-premises, for example you want private access from your on-premise data center
upvoted 2 times

Steve_4542636 1 year, 9 months ago

The RESTful services is neither an S3 or DynamDB service, so a VPC Gateway endpoint isn't available here.
upvoted 5 times

bdp123 1 year, 9 months ago

Selected Answer: B

fewest changes to code and below link:
https://gkzz.medium.com/what-is-the-differences-between-vpc-endpoint-gateway-endpoint-ae97bfab97d8

upvoted 3 times

PoisonBlack 1 year, 7 months ago

This really helped me understand the difference between the two. Thx

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #361 *Topic 1*

A company hosts a multiplayer gaming application on AWS. The company wants the application to read data with sub-millisecond latency and run one-time queries on historical data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon RDS for data that is frequently accessed. Run a periodic custom script to export the data to an Amazon S3 bucket.

B. Store the data directly in an Amazon S3 bucket. Implement an S3 Lifecycle policy to move older data to S3 Glacier Deep Archive for long-term storage. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

C. Use Amazon DynamoDB with DynamoDB Accelerator (DAX) for data that is frequently accessed. Export the data to an Amazon S3 bucket by using DynamoDB table export. Run one-time queries on the data in Amazon S3 by using Amazon Athena. **Most Voted**

D. Use Amazon DynamoDB for data that is frequently accessed. Turn on streaming to Amazon Kinesis Data Streams. Use Amazon Kinesis Data Firehose to read the data from Kinesis Data Streams. Store the records in an Amazon S3 bucket.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

lexotan Highly Voted 2 years, 1 month ago

Selected Answer: C

would be nice to have an explanation on why examtopic selects its answers.

upvoted 12 times

ale_brd_111 1 year, 5 months ago

exam topic does not select anything, these are questions from the free forum topics, the only thing exam topic does is to aggregate them all under one single point of view and if you pay you get to see them all aggregated else you can still scroll topic by topic for free

upvoted 7 times

TariqKipkemei Highly Voted 1 year, 7 months ago

Selected Answer: C

DAX delivers up to a 10 times performance improvement—from milliseconds to microseconds.
Using DynamoDB export to S3, you can export data from an Amazon DynamoDB table to an Amazon S3 bucket. This feature enables you to perform analytics and complex queries on your data using other AWS services such as Athena, AWS Glue.
upvoted 9 times

Omshanti Most Recent 8 months, 2 weeks ago

Selected Answer: C

test test
upvoted 2 times

sandordini 1 year, 1 month ago

Selected Answer: C

Sub-millisecond: DynamoDB (DAX), onetime query, Least operational overhead: Athena
upvoted 4 times

Uzbekistan 1 year, 3 months ago

Selected Answer: C

Dynamo DB + DAX = low latency.
upvoted 6 times

fabiomarrocolo 1 year, 3 months ago

Scusate io ho pagato contributor perchè vedo ancora + votati invece di vedere solo la risposta corretta? Grazie.Fabio
upvoted 2 times

LoXoL 1 year, 3 months ago

Vedrai sempre e comunque sia la risposta della community ("Most Voted") che la risposta degli admin (rettangolo verde).
Occhio perche' la risposta degli admin non sempre e' corretta.
upvoted 1 times

Mikado211 1 year, 6 months ago

Sub-millisecond latency == DAX
upvoted 4 times

Mikado211 1 year, 6 months ago

So C !
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

Amazon DynamoDB with DynamoDB Accelerator (DAX) is a fully managed, in-memory caching solution for DynamoDB. DAX can improve the performance of DynamoDB by up to 10x. This makes it a good choice for data that needs to be accessed with sub-millisecond latency.
DynamoDB table export allows you to export data from DynamoDB to an S3 bucket. This can be useful for running one-time queries on historical data.
Amazon Athena is a serverless, interactive query service that makes it easy to analyze data in Amazon S3. Athena can be used to run one-time queries on the data in the S3 bucket.
upvoted 6 times

aaroncelestin 1 year, 9 months ago

A NoSQL isn't even mentioned in the question and yet we are supposed to just imagine this fictional customer is using a NoSql DB
upvoted 3 times

marufxplorer 1 year, 11 months ago

C
Amazon DynamoDB with DynamoDB Accelerator (DAX): DynamoDB is a fully managed NoSQL database service provided by

AWS. It is designed for low-latency access to frequently accessed data. DynamoDB Accelerator (DAX) is an in-memory cache for DynamoDB that can significantly reduce read latency, making it suitable for achieving sub-millisecond read times.
upvoted 4 times

lucdt4 2 years ago

Selected Answer: C

C is correct
A don't meets a requirement (LEAST operational overhead) because use script
B: Not regarding to require
D: Kinesis for near-real-time (Not for read)
-> C is correct
upvoted 3 times

DagsH 2 years, 2 months ago

Selected Answer: C

Agreed C will be best because of DynamoDB DAX
upvoted 2 times

BeeKayEnn 2 years, 2 months ago

Option C will be the best fit.
As they would like to retrieve the data with sub-millisecond, DynamoDB with DAX is the answer.
DynamoDB supports some of the world's largest scale applications by providing consistent, single-digit millisecond response times at any scale. You can build applications with virtually unlimited throughput and storage.
upvoted 3 times

Grace83 2 years, 2 months ago

C is the correct answer
upvoted 2 times

KAUS2 2 years, 2 months ago

Selected Answer: C

Option C is the right one. The questions clearly states "sub-millisecond latency "
upvoted 3 times

smgsi 2 years, 2 months ago

Selected Answer: C

https://aws.amazon.com/dynamodb/dax/?nc1=h_ls
upvoted 4 times

[Removed] 2 years, 3 months ago

Selected Answer: C

Cccccccccccc
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #362 *Topic 1*

A company uses a payment processing system that requires messages for a particular payment ID to be received in the same order that they were sent. Otherwise, the payments might be processed incorrectly.

Which actions should a solutions architect take to meet this requirement? (Choose two.)

A. Write the messages to an Amazon DynamoDB table with the payment ID as the partition key.

B. Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key. **Most Voted**

C. Write the messages to an Amazon ElastiCache for Memcached cluster with the payment ID as the key.

D. Write the messages to an Amazon Simple Queue Service (Amazon SQS) queue. Set the message attribute to use the payment ID.

E. Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID. **Most Voted**

Correct Answer: *BE*

Community vote distribution

BE (80%) AE (17%) DE

Comments

Ashkan_10 Highly Voted 2 years, 2 months ago

Selected Answer: BE

Option B is preferred over A because Amazon Kinesis Data Streams inherently maintain the order of records within a shard, which is crucial for the given requirement of preserving the order of messages for a particular payment ID. When you use the payment ID as the partition key, all messages for that payment ID will be sent to the same shard, ensuring that the order of messages is maintained.

On the other hand, Amazon DynamoDB is a NoSQL database service that provides fast and predictable performance with seamless scalability. While it can store data with partition keys, it does not guarantee the order of records within a partition, which is essential for the given use case. Hence, using Kinesis Data Streams is more suitable for this requirement.

As DynamoDB does not keep the order, I think BE is the correct answer here.

upvoted 31 times

awsgeek75 Highly Voted 1 year, 4 months ago

I don't understand the question. The only requirement is: " system that requires messages for a particular payment ID to be received in the same order that they were sent"

SQS FIFO (E) meets this requirement.

Why would you "write the message" to Kinesis or DynamoDB anymore. There is no streaming or DB storage requirement in the question. Between A/B, B is better logically but it doesn't meet any stated requirement.

Happy to understand what I'm missing

upvoted 9 times

MatAlves 9 months ago

Instead of "what actions...", the question should say "what are the alternatives/options that meet this requirement".

upvoted 4 times

Manimgh Most Recent 1 month, 3 weeks ago

Selected Answer: BE

same order = FIFO first in first out

upvoted 1 times

FlyingHawk 6 months, 1 week ago

Selected Answer: BE

Both B and E can ensure the messages being processed in the order they were received, however SQS FIFO queues (E) are simpler, more cost-effective, and better aligned with the needs of this use case compared to Kinesis. Unless you have additional real-time analytics or high-throughput requirements, E is the superior choice, the big reason why KDS can preserve the order is due to sequence number assigned to data record when it was written to shard, Dynamo DB does not have a sequence number associated with the record. https://docs.aws.amazon.com/streams/latest/dev/key-concepts.html

upvoted 2 times

pentium75 1 year, 5 months ago

Selected Answer: BE

Both Kinesis and SQS FIFO queue guarantee the order, other answers don't.

upvoted 4 times

meowruki 1 year, 6 months ago

Option B (Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key): Kinesis can provide ordered processing within a shard
Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID.

SQS FIFO (First-In-First-Out) queues preserve the order of messages within a message group.

upvoted 4 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: BE

Technically both B and E will ensure processing order, but SQS FIFO was specifically built to handle this requirement.
There is no ask on how to store the data so A and C are out.

upvoted 2 times

Pritam228 1 year, 8 months ago

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.Partitions.html

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: DE

options D and E are better because they mimic a real-world queue system and ensure that payments are processed in the correct order, just like customers in a store would be served in the order they arrived. This is crucial for a payment processing system where order matters to avoid mistakes in payment processing.

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Amazon Kinesis Data Streams Overkill for Ordering
Overkill for Ordering: While Kinesis can maintain order within a partition key, it might be seen as overkill for a scenario where your primary concern is maintaining the order of payments. SQS FIFO queues (option E) are specifically designed for this purpose and provide an easier and more cost-effective solution.
upvoted 1 times

omoakin 2 years ago

AAAAAAAAA EEEEEEEEEEEEE
upvoted 3 times

Konb 2 years ago

Selected Answer: AE

IF the question would be "Choose all the solutions that fulfill these requirements" I would chosen BE.

But it is:
"Which actions should a solutions architect take to meet this requirement? "

For this reason I chose AE, because we don't need both Kinesis AND SQS for this solution. Both choices complement to order processing: order stored in DB, work item goes to the queue.
upvoted 3 times

Smart 1 year, 10 months ago

Incorrect, AWS will clarify it by using the phrase - "combination of actions".
upvoted 2 times

luisgu 2 years ago

Selected Answer: BE

E --> no doubt
B --> see https://docs.aws.amazon.com/streams/latest/dev/key-concepts.html
upvoted 2 times

kruasan 2 years, 1 month ago

Selected Answer: BE

1) SQS FIFO queues guarantee that messages are received in the exact order they are sent. Using the payment ID as the message group ensures all messages for a payment ID are received sequentially.
2) Kinesis data streams can also enforce ordering on a per partition key basis. Using the payment ID as the partition key will ensure strict ordering of messages for each payment ID.
upvoted 3 times

kruasan 2 years, 1 month ago

The other options do not guarantee message ordering. DynamoDB and ElastiCache are not message queues. SQS standard queues deliver messages in approximate order only.
upvoted 3 times

mrgeee 2 years, 1 month ago

Selected Answer: BE

BE no doubt.
upvoted 2 times

nosense 2 years, 1 month ago

Selected Answer: BE

Option A, writing the messages to an Amazon DynamoDB table, would not necessarily preserve the order of messages for a particular payment ID
upvoted 2 times

MssP 2 years, 2 months ago

Selected Answer: BE

I don´t unsderstand A. How you can guaratee the order with DynamoDB?? The order is guarantee with SQS FIFO and Kinesis

I don't understand A, How you can guarantee the order with DynamoDB? The order is guarantee with SQS FIFO and Kinesis Data Stream in 1 shard...

upvoted 5 times

pentium75 1 year, 5 months ago

If it really means "combination of actions" than A+E would work, because you'd use the FIFO queue (E) to guarantee the order. Then the order in the database doesn't matter. If they want to alternative solutions then obviously B and E would work while A alone doesn't.

upvoted 2 times

Grace83 2 years, 2 months ago

AE is the answer

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #363 *Topic 1*

A company is building a game system that needs to send unique events to separate leaderboard, matchmaking, and authentication services concurrently. The company needs an AWS event-driven system that guarantees the order of the events.

Which solution will meet these requirements?

A. Amazon EventBridge event bus
B. Amazon Simple Notification Service (Amazon SNS) FIFO topics **Most Voted**
C. Amazon Simple Notification Service (Amazon SNS) standard topics
D. Amazon Simple Queue Service (Amazon SQS) FIFO queues

Correct Answer: *B*

Community vote distribution

B (79%) D (16%) A (5%)

Comments

bella Highly Voted 2 years, 1 month ago

Selected Answer: B

I don't honestly / can't understand why people go to ChapGPT to ask for the answers.... if I recall correctly they only consolidated their DB until 2021...

upvoted 18 times

LeonSauveterre 6 months, 1 week ago

Yes, it used to be just like that. Fortunately things have changed as of today - Nov 29 2024 in my timezone. Personally I find ChatGPT + Amazon Q to be quite awesome for SAA tests

upvoted 2 times

aaroncelestin 1 year, 9 months ago

Yup, ChatGPT doesn't //know// anything about AWS services. It only repeats what other people have said about it, which could be nonsense or hyperbole or some combination thereof.

upvoted 5 times

LazyTs Highly Voted 1 year, 9 months ago

Selected Answer: B

The answer is B la. SNS FIFO topics queue should be used combined with SQS FIFO queue in this case. The question asked for correct order to different event, so asking for SNS fan out here to send to individual SQS.
https://docs.aws.amazon.com/sns/latest/dg/fifo-example-use-case.html
upvoted 14 times

dkw2342 1 year, 3 months ago

B is correct, but this is not about SNS -> SQS fan-out, it's not necessary. Just SNS FIFO for ordered pub/sub messaging.
upvoted 3 times

Po_chih 1 year, 8 months ago

The best answer!
upvoted 1 times

FlyingHawk Most Recent 6 months, 1 week ago

Selected Answer: B

Base on this doc https://docs.aws.amazon.com/sns/latest/dg/fifo-topic-message-ordering.html, the answer should B and D, however, if you can only select one, then B.
upvoted 1 times

MatAlves 9 months ago

Selected Answer: B

SNS can have many-to-many relations, while SQS supports only one consumer at a time (many-to-one).
upvoted 5 times

1e22522 10 months, 1 week ago

First time in my life that the answer is actually SNS
upvoted 5 times

richiexamaws 1 year ago

Selected Answer: D

AWS does not currently offer FIFO topics for SNS. SNS only supports standard topics, which do not guarantee message order.
upvoted 1 times

elmyth 9 months ago

I'm creating a topic right now and I have both types to choose.
upvoted 3 times

Darshan07 1 year, 3 months ago

Selected Answer: B

Even chat gpt said B
upvoted 2 times

awsgeek75 1 year, 4 months ago

Selected Answer: B

Yes, you can technically do this with SQS FIFO partitioned queue by giving separate group ID's to leaderboard, matchmaking etc but this is not as useful as SNS FIFO and is overkill as no need for storage etc. B is more elegant and concise solution,
upvoted 5 times

foha2012 1 year, 5 months ago

Guys, ChatGPT sucks !. Try removing [most voted] from choice B and it will choose D. And if you put [most voted] in front of A, it will select A. LOL !
upvoted 3 times

Marco_St 1 year, 5 months ago

Selected Answer: B

iust know SNS FIFO also can send events or messages cocurrently to many subscribers while maintaining the order it receives

just know SNS FIFO also can send events or messages cocurrently to many subscribers while maintaining the order it receives.
SNS fanout pattern is set in standard SNS which is commonly used to fan out events to large number of subscribers and usually for duplicated messages.

upvoted 2 times

Mikado211 1 year, 6 months ago

Selected Answer: B

SQS looks like a good idea first, but since we have to send the same message to multiple destination, even if SQS could do it, SNS is much more dedicated to this kind of usage.

upvoted 5 times

sparun1607 1 year, 6 months ago

My Answer is B

https://docs.aws.amazon.com/sns/latest/dg/sns-fifo-topics.html

You can use Amazon SNS FIFO (first in, first out) topics with Amazon SQS FIFO queues to provide strict message ordering and message deduplication. The FIFO capabilities of each of these services work together to act as a fully managed service to integrate distributed applications that require data consistency in near-real time. Subscribing Amazon SQS standard queues to Amazon SNS FIFO topics provides best-effort ordering and at least once delivery.

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

bbbbbbbbbbbbbb

upvoted 2 times

jaydesai8 1 year, 11 months ago

Selected Answer: D

SQS FIFO maintains the order of the events - Answer is D

upvoted 2 times

jayce5 2 years ago

Selected Answer: B

It should be the fan-out pattern, and the pattern starts with Amazon SNS FIFO for the orders.

upvoted 3 times

danielklein09 2 years ago

Selected Answer: D

Answer is D. You are so lazy because instead of searching in documentation or your notes, you are asking ChatGPT. Do you really think you will take this exam ? Hint: ask ChatGPT

upvoted 5 times

lucdt4 2 years ago

Selected Answer: D

D is correct (SQS FIFO)
Because B can't send event concurrently though it can send in the order of the events

upvoted 1 times